Tensor-Based Robust Principal Component Analysis With Locality Preserving Graph and Frontal Slice Sparsity for Hyperspectral Image Classification

Yingxu Wang, Tianjun Li, Long Chen, *Member, IEEE*, Yufeng Yu, Yinping Zhao, and Jin Zhou, *Member, IEEE*

***Abstract*— Tensor-based robust principal component analysis (PCA) methods are efficient to discover the low-rank part of a hyperspectral image for reducing redundant information and guarantee good classification results. However, current methods cannot remove noise adequately, and the residual noise remaining in the low-rank image limits the further improvement of classification performance. Thus, enhancing the robustness to noise is important and helpful for tensor-based robust PCA (RPCA) methods to process hyperspectral images. To this end, we propose a tensor-based RPCA method with a locality preserving graph and frontal slice sparsity (LPGTRPCA) for hyperspectral image classification. Specifically, a tensor $l_{2,2,1}$ norm that requires the frontal slice sparsity of a tensor is defined to extract the noise in the hyperspectral image from the frontal direction. What is more, a position-based Laplacian graph that preserves the local structures of a tensor according to the spatial position is designed for relieving the impact of the residual noise remaining in the low-rank image. Based on the tensor nuclear norm, the tensor $l_{2,2,1}$ norm, and the position-based Laplacian graph, LPGTRPCA efficiently separates the low-rank part with little noise from a raw hyperspectral image and achieves more robust classification results than current methods. LPGTRPCA is optimized by the alternative direction multiplier method (ADMM), and the convergence of solutions is experimentally demonstrated. In the experiments conducted on Indian Pines, Pavia University, and Salinas datasets, LPGTRPCA outperformed various state-of-the-art and classical tensor-based RPCA methods in terms of average class classification accuracy (AA), overall classification accuracy (OA), and kappa coefficient (KC).**

***Index Terms*— Frontal slice sparsity, hyperspectral image classification, locality preserving graph, tensor-based robust principal component analysis (PCA).**

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Yingxu Wang, Long Chen, and Yufeng Yu are with the Department of Computer and Information Science, Faculty of Science and Technology, University of Macau, Macau, China (e-mail: longchen@um.edu.mo).

Tianjun Li is with the School of Computer Science and Engineering, South China University of Technology, Guangzhou 510641, China.

Yinping Zhao is with the School of Software, Northwestern Polytechnical University, Xi'an 710072, China.

Jin Zhou is with the Shandong Provincial Key Laboratory of Network Based Intelligent Computing, University of Jinan, Jinan 250022, China.

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I. Introduction

THE continuous narrow spectral bands of hyperspectral images contain abundant information for target recognition. However, the huge number of spectral bands and sparse noise hinders their convenient utilization. Thus preprocessing is required before their application. Processing methods can be divided into categories, such as unmixing [1]–[3], denoising [4]–[6], band selection [7], [8], and feature extraction or dimension reduction [9], [10]. The principal component analysis (PCA) is an unsupervised method used in various hyperspectral image processing models to discover the main principal components or reduce redundancies. Tsagaris *et al.* [11] proposed the use of PCA to extract the first principal component of each subgroup of hyperspectral bands for visualization. A similar method with 2-D PCA was proposed to avoid the destruction of row-to-row relationships [12]. Chen *et al.* [13] proposed a hyperspectral image denoising method based on PCA and block-matching 4-D (BM4D), regarding the first few principal components as fine features without noise and the rest as denoised by BM4D. Xia *et al.* [14] applied a hyperspectral image classification method with rotation random forest via kernel PCA (KPCA) to randomly split original feature subsets to discover high-order features for input to a random forest. Kang *et al.* [15] employed PCA to reduce the spectral dimension of stacked edge-preserving features to represent features in the mean square sense and highlight the separable degree of pixels and obtained good classification results of hyperspectral images. Jiang *et al.* [16] performed PCA on the homogeneous region gained from superpixel partition to extract inherent low-dimensional features of hyperspectral images. Kang *et al.* [17] proposed hyperspectral image visualization using PCA to fuse base layers obtained by edge-preserving filtering.

The standard PCA method is sensitive to outliers [18], [19], so the performance of the above PCA-based methods may be influenced by noise in hyperspectral data. To achieve insensitivity to noise, many researchers have proposed to extract the main component of an estimated robust covariant matrix instead of the original covariant matrix of the data [18]–[20]. These methods have limitations, such as performance deterioration with an increase in the data dimension and high computational complexity. Robust PCA (RPCA) based on outlier pursuit [21] has been proposed to decompose a noisy

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2-D matrix into a low-rank clean matrix and a sparse error matrix, using the nuclear and l_1 norms, respectively, to ensure the low-rank property and sparsity of matrices. This method is solved by the augmented Lagrange multiplier (ALM) algorithm [21] in polynomial time complexity, and its performance is independent of the data dimension. Xu *et al.* [22] proposed using the $l_{2,1}$ norm to restrict the column sparsity of the error matrix and proposed columnwise sparse RPCA (CWRPCA).

RPCA and its variants have achieved great success in hyperspectral image processing tasks. Cao *et al.* [23] proposed a robust restoration method by simultaneously discovering the spatial–spectral low-rank characterization of the clean hyperspectral image and the intrinsic structures of the mixed noise in the sparse error image. Sun and Du [24] incorporated a Laplacian graph term in RPCA to preserve the smooth manifold of the band set in the low-rank clean part of a hyperspectral image for further band selection. At the same time, Sun *et al.* [25] also proposed a compressed columnwise RPCA method for hyperspectral anomaly detection, in which Hadamard random projection was used to decrease the computational load, and the $l_{2,1}$ norm was employed to reduce the influence of anomalies in the background. Zu *et al.* [26] applied the CWRPCA model to the homogeneous regions of hyperspectral images obtained by simple linear iterative clustering, so as to simultaneously capture the spectral and spatial information. Rambhatla *et al.* [27] proposed a dictionary-based RPCA in which the decomposed error matrix was sparse in a given dictionary and applied this model to localize targets in hyperspectral images under the guidance of a dictionary containing spectral features of interesting structures [28]. They further improved their target localization method by considering both entrywise and columnwise sparsities to discover different structures of dictionary sparse components [29]. The data of a hyperspectral image form a 3-D tensor instead of a 2-D matrix, so data reshaping is required before conducting hyperspectral image processing methods based on RPCA or CWRPCA. However, this may destroy the inherent structures of images and limit the final results. A good solution is to avoid data deformation by applying tensor-based models rather than matrix-based models.

Tensor-based RPCA decomposes a tensor of three or more dimensions into low-rank and sparse noisy parts. Various definitions have been proposed to characterize a tensor's rank, such as the Tucker rank [30], the tensor multirank [31], and the tensor tubal rank [32]. Several convex relaxations of the tensor rank are available for easy practical application. Based on different constraints of tensor rank and sparsity, many tensor-based RPCA methods have been proposed. Kilmer *et al.* [31] and Kilmer and Martin [33] designed the first tensor nuclear norm as the tight relaxation of the tensor multirank. Based on this, they proposed a tensor-based RPCA method (TRPCA14) with tubewise sparse constricted by tensor $l_{1,1,2}$ norm for multilinear imaging recovery [32]. Ren *et al.* [34] presented tensor outlier pursuit with the relaxation of the Tucker rank and proposed a tensor-based RPCA method (FWTRPCA) to discover the framewise outliers in a tensor. Lu *et al.* [35] studied the recovery guarantee of a tensor-based RPCA model (TRPCA16), using a modified tensor nuclear norm and a tensor l_1 norm to constrain the low-rank property and sparsity of a tensor. Based on the tensor nuclear norm [36] and the tensor $l_{2,1,2}$ norm, Zhou and Feng [36] proposed outlier robust tensor-based PCA (ORTPCA) under the assumption that outliers are distributed along the second dimension. Liu *et al.* [37] defined a tensor nuclear norm with a low-rank core matrix to further discover the low-rank structures hidden in a tensor and proposed an improved robust tensor-based PCA (IRTPCA) method. Lu *et al.* [38] proposed a new tensor nuclear norm and proved the recovery guarantee of the tensor-based RPCA problem (TRPCA19) to be the extension of that of RPCA. Sun *et al.* [39] proposed lateral slice sparse tensor-based RPCA (LSSTRPCA) to remove outliers distributed in lateral slices for hyperspectral image classification, using the tensor nuclear norm [35], [38] to characterize the low-rank structures of the clean part of a hyperspectral image and the $l_{2,1,2}$ norm to ensure the sparsity of each lateral slice in the noise part. Benefiting from the good properties of tensor factorization, tensor-based RPCA methods can better extract the low-rank clean part of hyperspectral images than matrix-based RPCA and CWRPCA. However, such methods cannot adequately remove all sparse outliers, and residual noise destroys the locality of the classification results of the clean image.

Thus, enhancing the ability to relieve the impact of noise is important and helpful for tensor-based RPCA to process hyperspectral images. To this end, we propose a tensor-based RPCA method with a locality preserving graph and frontal slice sparsity (LPGTRPCA) for hyperspectral image classification. In this article, the tensor $l_{2,2,1}$ norm is defined for constraining the frontal slice sparsity of a tensor, and a position-based Laplacian graph is designed to preserve the local structures. Our motivations are to use the tensor-based RPCA model with $l_{2,2,1}$ norm to separate the low-rank part and the noise part of a hyperspectral image and use the position-based Laplacian graph to further relieve the impact of the residual noise in the low-rank part. The $l_{2,2,1}$ norm and the position-based locality preserving graph work together to guarantee the good performance of LPGTRPCA.

On the one hand, considering that sparse noise exists in a small subset of spectral bands or some local areas of a hyperspectral image [39], the proposed $l_{2,2,1}$ norm is more suitable for sparse noise excavation than other tensor sparsity norms. The $l_{1,1,2}$ norm [32] extracts the tubewise sparse noise in multilinear images but is not applicable to noise distributed in hyperspectral images. Similarly, the l_1 norm [35], [38] focusing on the global sparsity of a tensor performs poorly on hyperspectral image classification [39]. The $l_{2,1,2}$ norm constricting the lateral slice sparsity [36], [39] behaves better than the l_1 norm and the $l_{1,1,2}$ norm, but it may result in uneven locality and, thus, limit the further improvement of classification results.

On the other hand, compared to other graph learning models, the position-based Laplacian graph is more suitable for locality preservation in hyperspectral image processing. Ma *et al.* [40] proposed a local-manifold-learning-based graph

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to classify hyperspectral images with complex geometry and multiple submanifolds. Sellars *et al.* [41] used the superpixel technique in the graph representation to accelerate the classification of hyperspectral images. He *et al.* [42] proposed a parameter-free anchor graph with low computational complexity for large hyperspectral images. Luo *et al.* [43] proposed a hybrid graph learning method to discover the inherent high-order relationships between samples of a hyperspectral image. The above methods are semisupervised or supervised, with graphs constructed based on labeled data. Similar work using label-based graphs has been found in [44]–[46]. Besides, a few methods have introduced intensity-based graphs to the RPCA model. Sun *et al.* [25] incorporated a Laplacian graph term in RPCA to preserve the smooth manifold of the band set in the low-rank clean part of a hyperspectral image for further band selection. Nie *et al.* [47] used spectral graph regularization to preserve geometric structures for hyperspectral image denoising (GTRPCA). Both label- and intensity-based graph regularizations have innate defects. The construction of label-based graphs requires labeled data, which are costly and time-consuming to acquire. Intensity-based graphs are sensitive to noise, which is difficult to thoroughly remove from hyperspectral images. In contrast, position-based graph regularization does not rely on labeled data and is robust to noise. The contributions of this study are given as follows.

1) The tensor $l_{2,2,1}$ norm is defined to constrain the frontal slice sparsity of a tensor. The tensor $l_{2,2,1}$ norm is more suitable to handle hyperspectral images than other tensor sparse norms.
2) A position-based Laplacian graph is designed to preserve the local structures for a hyperspectral image to relieve the impact of noise.
3) An LPGTRPCA method is proposed for hyperspectral image classification. This method efficiently separates the low-rank part from a raw hyperspectral image and achieves robust classification results.

The rest of this article is organized as follows. Preliminaries and related work are provided in Section II. LPGTRPCA is proposed in Section III. Experimental results are analyzed in Section IV. Conclusions are given in Section V.

II. Preliminaries and Related Work

A. Notations and Tensor Preliminaries

In this article, 3-D tensors are denoted by calligraphy capital letters. For example, $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is a tensor with size of $n_1 \times n_2 \times n_3$. The ith horizontal slice, the lateral slice, and the frontal slice of $\mathcal{A}$ are denoted as $\mathcal{A}(i,:,:)$, $\mathcal{A}(:,i,:)$, and $\mathcal{A}(:,:,i)$. The entry of $\mathcal{A}$ is denoted as $\mathcal{A}(i,j,k)$. $\overline{\mathcal{A}} = \text{fft}(\mathcal{A},[],:)$ is the discrete Fourier transformation of $\mathcal{A}$ among the third dimension [31], [33], [38]. Correspondingly, $\overline{\mathcal{A}}$ can be converted back to $\mathcal{A}$ by using inverse discrete Fourier transformation, i.e., $\mathcal{A} = \text{ifft}(\overline{\mathcal{A}},[],3)$. The 2-D matrices are denoted by normal capital letters, such as $A \in \mathbb{R}^{n_1 \times n_2}$. Lowercase letters denote scalars. For the matrix A, it can be decomposed into three matrices by singular value decomposition (SVD) [31], [33], [38] as

$$A = USV^T \tag{1}$$

where $U \in \mathbb{R}^{n_1 \times n_1}$, $S \in \mathbb{R}^{n_1 \times n_2}$, and $V \in \mathbb{R}^{n_2 \times n_2}$.

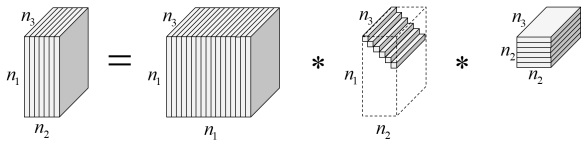


Fig. 1. t-SVD of an $n_1 \times n_2 \times n_3$ tensor.

Similarly, a tensor $\mathcal{A}$ can be decomposed into three tensors by tensor SVD (t-SVD) [31], [33], [38] as

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{2}$$

where $*$ is the tensor product (t-product) [31], [33], [38]. $\mathcal{U} \in \mathbb{C}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{C}^{n_2 \times n_2 \times n_3}$ are two orthogonal tensors. $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f-diagonal tensor. Fig. 1 illustrates the t-SVD [31], [33], [38] of an $n_1 \times n_2 \times n_3$ tensor.

$\mathcal{U}$, $\mathcal{S}$, and $\mathcal{V}$ of tensor $\mathcal{A}$ can be calculated by conducting matrix SVD in the Fourier domain [31], [33], [38]. Based on t-SVD, the tensor tubal rank [31], [33] was proposed to measure the complexity of a tensor.

Definition 1 (Tensor Tubal Rank [31], [33]): The tensor tubal rank $\text{Rank}_t(\mathcal{A})$ of tensor $\mathcal{A}$ is the number of nonzero tubes of $\mathcal{S}$, where $\mathcal{S}$ is obtained from $\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$.

The tensor nuclear norm [31], [33], [35], [38] was designed as the convex relaxation of tensor tubal rank. In this article, $\| \ \|_*$ denotes the nuclear norm of a tensor or a matrix. Specifically, $\|\mathcal{A}\|_*$ is the tensor nuclear norm of $\mathcal{A}$, and $\|A\|_* = \sum_{i=1}^{\min(n_1,n_2)} \sigma_i(A)$ is the matrix nuclear norm [19] of A, where $\sigma_i(A)$ is the ith singular value of A. There are several different definitions of tensor nuclear norm as described in Definitions 2–4.

Definition 2 (Tensor Nuclear Norm): First, Kilmer *et al.* [31] and Kilmer and Martin [33] define the tensor nuclear norm $\|\mathcal{A}\|_*$ as the sum of matrix nuclear norm of all frontal slices of $\overline{\mathcal{A}}$ as follows:

$$\|\mathcal{A}\|_* = \sum_{i=1}^{n_3} \|\overline{\mathcal{A}}(:,:,i)\|_*. \tag{3}$$

Definition 3 (Tensor Nuclear Norm): Then, Lu *et al.* [35] define the tensor nuclear norm $\|\mathcal{A}\|_*$ as the average of matrix nuclear norm of all frontal slices of $\overline{\mathcal{A}}$, as shown follows:

$$\|\mathcal{A}\|_* = \frac{1}{n_3} \sum_{i=1}^{n_3} \|\overline{\mathcal{A}}(:,:,i)\|_*. \tag{4}$$

Definition 4 (Tensor Nuclear Norm): Recently, Lu *et al.* [38] redefine the tensor nuclear norm as

$$\|\mathcal{A}\|_* = \sum_{i=1}^{\text{Rank}_t(\mathcal{A})} \mathcal{S}(i,i,1) \tag{5}$$

where $\text{Rank}_t(\mathcal{A})$ is the tubal rank of $\mathcal{A}$.

Using the tensor nuclear norm to recover low-complexity tensors results in the following optimization problem:

$$\min_{\mathcal{A}} \ \tau\|\mathcal{A}\|_* + \frac{1}{2}\|\mathcal{A} - \mathcal{B}\|_F^2 \tag{6}$$

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Algorithm 1 Tensor-SVT [38]

Input: Tensor $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, scalar $\tau \geq 0$.
Output: $\Gamma_\tau(\mathcal{B})$.
1. Calculate $\overline{\mathcal{B}} = fft(\mathcal{B}, [], 3)$.
2. Conduct matrix SVT on first $\lceil \frac{n_3+1}{2} \rceil$ frontal slices of $\overline{\mathcal{B}}$.
   for $i = 1, \ldots, \lceil \frac{n_3+1}{2} \rceil$, **do**
   $[U, S, V] = \text{SVD}(\overline{\mathcal{B}}(:, :, i))$;
   $\overline{\mathcal{D}}(:, :, i) = U(S - \tau)_+ V^T$;
   end for
3. Calculate rest slices of $\overline{\mathcal{D}}$.
   for $i = \lceil \frac{n_3+1}{2} \rceil + 1, \ldots, n_3$, **do**
   $\overline{\mathcal{D}}(:, :, i) = conj(\overline{\mathcal{D}}(:, :, n_3 - i + 2))$
   end for
4. Calculate $\Gamma_\tau(\mathcal{B}) = ifft(\overline{\mathcal{D}}, [], 3)$.

where $\mathcal{B}$ is a tensor with same size as $\mathcal{A}$, and $\| \ \|_F$ denotes the Frobenius norm of a tensor, i.e., $\|\mathcal{A}\|_F = (\sum_{i,j,k} \|\mathcal{A}(i, j, k)\|^2)^{1/2}$.

The researches in [33] and [38] proposed to solve (6) after breaking up it in to n_3 independent minimization problems corresponding to each frontal slice, and the final results were given as

$$\Gamma_\tau(\mathcal{B}) = \underset{\mathcal{A}}{\text{argmin}} \ \tau\|\mathcal{A}\|_* + \frac{1}{2}\|\mathcal{A} - \mathcal{B}\|_F^2. \tag{7}$$

There are two different methods to calculate $\Gamma_\tau(\mathcal{B})$. Specifically, if Definition 1 or 2 is used, $\Gamma_\tau(\mathcal{B})$ is calculated according to [33]. If Definition 3 is used, $\Gamma_\tau(\mathcal{B})$ is calculated by tensor singular value thresholding (tensor-SVT) proposed in [38], as shown in Algorithm 1.

B. Tensor-Based RPCA

The tensor-based RPCA [32], [34]–[39] aimed to decompose a noisy tensor into a low-rank tensor $\mathcal{L}$ and a sparse error tensor $\mathcal{E}$. The general model of tensor-based RPCA was designed as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}} \ \text{Rank}(\mathcal{L}) + \alpha\text{Density}(\mathcal{E}) \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E} \end{aligned} \tag{8}$$

where the first term denoted the rank or corresponding relaxations of the tensor $\mathcal{L}$ and the second term denoted the density of the nonzero entries in tensor $\mathcal{E}$. α was a tradeoff parameter. Minimizing this objective function required the low-rank properties of $\mathcal{L}$ and the sparsity of $\mathcal{E}$ simultaneously.

In the literature, various techniques were incorporated into this model. For the first term, the sum of nuclear norms [48], [49] and different tensor nuclear norms [32], [35], [38] were qualified to ensure the low-rank properties. For the second term, as shown in Fig. 2, the tensor l_1 norm [35], [38] and mixed norms, such as the tensor $l_{1,1,2}$ norm [32] and the tensor $l_{2,1,2}$ norm [39], were used to ensure the tensor sparsity. The comparison between various tensor-based RPCA models is provided in Section III.

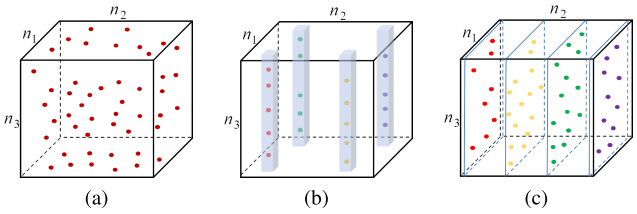


Fig. 2. Performance of various tensor sparse norms. (a) l_1 norm for global sparsity. (b) $l_{1,1,2}$ norm for tubewise sparsity. (c) $l_{2,1,2}$ norm for lateral slice sparsity.

III. Tensor-Based Robust Principal Component Analysis With Locality Preserving Graph for Hyperspectral Image Classification

Tensor-based RPCA methods can separate the low-rank part from a hyperspectral image for classification. However, current methods cannot adequately remove all sparse outliers, and the residual noise limits the classification performance. To solve this issue, we propose the LPGTRPCA for more robust classification results. In this section, the tensor $l_{2,2,1}$ norm that constrains the frontal slice sparsity of a tensor and a position-based graph for preserving the local structures according to the spatial position is first designed. Subsequently, LPGTRPCA is established and then compared with other tensor-based RPCA methods. Finally, the solutions of LPGTRPCA are yielded by the alternative direction multiplier method (ADMM) [50].

A. Tensor-Based RPCA With Frontal Slice Sparsity

In the literature, the tensor $l_{2,1,2}$ norm [36], [39] that extracts outliers from lateral direction performed better than the tensor l_1 norm [38] and the tensor $l_{1,1,2}$ norm [32] on hyperspectral images because the noise usually exists in a small subset of spectral bands or some local areas [39]. However, the tensor $l_{2,1,2}$ norm may result in the uneven locality and, thus, limit the further improvement of classification results. In this part, we define the tensor $l_{2,2,1}$ norm to constrain the frontal slice sparsity and embed it into the tensor-based RPCA model as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}} \ \|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,2,1} \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E} \end{aligned} \tag{9}$$

where $\mathcal{X}$ is the 3-D tensor with size $n_1 * n_2 * n_3$. $\mathcal{L}$ is the tensor of the low-rank part, and $\|\mathcal{L}\|_*$ is the tensor nuclear norm [38] of $\mathcal{L}$. $\mathcal{E}$ is the tensor of sparse noise part, and $\|\mathcal{E}\|_{2,2,1} = \sum_{i=1}^{n_3} \|\mathcal{E}(:, :, i)\|_F$ is the $l_{2,2,1}$ norm of $\mathcal{E}$. α is a tradeoff parameter. Minimizing this objective function decomposes a corrupted tensor into a low-rank tensor and a sparse tensor with frontal slice sparsity, as shown in Fig. 3.

Both the tensor $l_{2,1,2}$ norm and the tensor $l_{2,2,1}$ norm are able to discover most of the sparse noise in hyperspectral images, but the new defined $l_{2,2,1}$ norm can lead to smoother locality in frontal direction. Specifically, the $l_{2,2,1}$ norm extracts sparse outliers for each frontal slice, and thus, the values of the pixels in a local area of the low-rank part are close to each other. As shown in Fig. 4(a), the low-rank part obtained by

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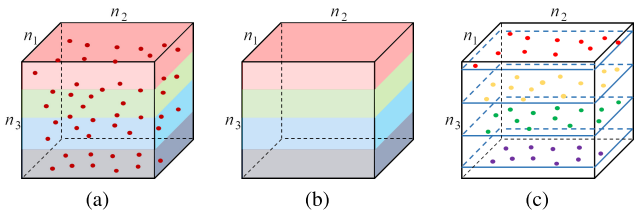


Fig. 3. Decompose (a) corrupted tensor into (b) low-rank tensor and (c) sparse tensor with frontal slice sparsity.

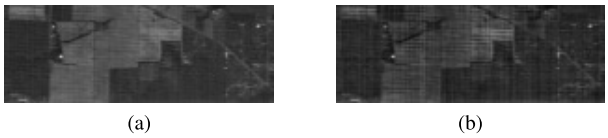


Fig. 4. Plots of Rows 65–116 of the first band of the low-rank image obtained by (a) proposed method and (b) LSSTRPCA on Indian Pines [51].

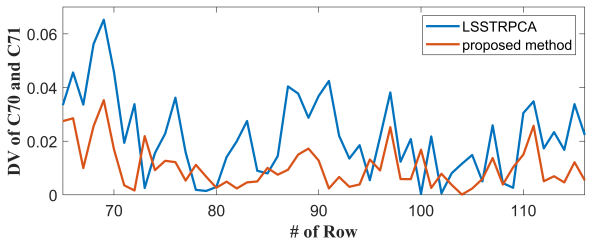


Fig. 5. DV of C70 and C71 of the first band of the low-rank image obtained by LSSTRPCA and the proposed method on Indian Pines [51].

the tensor-RPCA with the newly defined tensor $l_{2,2,1}$ norm is smooth. In contrast, the tensor $l_{2,1,2}$ norm used in previous work [36], [39] ensures the smoothness of each lateral slice but may result in the uneven locality in the frontal direction, especially when two successive lateral slices contain different level noise. As shown in Fig. 4(b), the first band of the low-rank part obtained by LSSTRPCA with $l_{2,1,2}$ norm is uneven. Fig. 5 plots the difference between the pixel values (DV) of Column 70 (C70) and Column 71 (C71) of the first band of the low-rank image obtained by LSSTRPCA and tensor-based RPCA with new defined $l_{2,2,1}$ norm. We can see that the DV obtained by our method is obviously smaller than the one of LSSTRPCA. In conclusion, the $l_{2,2,1}$ norm can lead to smoother locality in frontal direction than $l_{2,1,2}$ norm and, thus, is more suitable for hyperspectral images.

B. Position-Based Locality Preserving Graph for Hyperspectral Image

Practically, though tensor-based RPCA methods can remove most of the noise of the low-rank part $\mathcal{L}$ of a hyperspectral image, the residual noise still limits the classification results. To solve this issue, we design a position-based Laplacian graph to preserve local structures to relieve the impact of the residual noise. Specifically, the position-based Laplacian graph constrains each pixel to have a similar value with its neighbors, which are defined as the pixels located in a square window that is centralized at this pixel, as shown in Fig. 6. In this

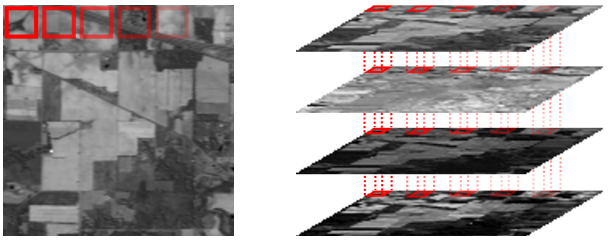


Fig. 6. Neighborhood of a pixel is defined as a square window centralized at itself. Observation of (a) single band and (b) multiple bands.

$$\begin{bmatrix} p-r*n_1-r & \dots & p-n_1-r & p-r & p+n_1-r & \dots & p+r*n_1-r \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ p-r*n_1-1 & \dots & p-n_1-1 & p-1 & p+n_1-1 & \dots & p+r*n_1-1 \\ p-r*n_1 & \dots & p-n_1 & p & p+n_1 & \dots & p+r*n_1 \\ p-r*n_1+1 & \dots & p-n_1+1 & p+1 & p+n_1+1 & \dots & p+r*n_1+1 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ p-r*n_1+r & \dots & p-n_1+r & p+r & p+n_1+r & \dots & p+r*n_1+r \end{bmatrix}$$

Fig. 7. Index of the $(2r+1)^2-1$ pixels that are located in the square window centralized at the pth pixel with radius r.

way, the values of noise pixels could be revised. It should be noted that the neighbor set for each pixel is constructed based on the spatial position, and thus, the local consistency of the original hyperspectral image is not destroyed.

To construct the position-based graph, we first reshape the 3-D low-rank part $\mathcal{L}$ into a 2-D matrix $\mathcal{L}_{(3)} \in \mathbb{R}^{n_3 \times (n_1\ n_2)}$ by tensor *mode-3* unfolding [52] as

$$\mathcal{L}_{(3)} = \text{unfold}(\mathcal{L}) = [\mathcal{L}(:,1,:)^T, \mathcal{L}(:,2,:)^T, \dots, \mathcal{L}(:,n_2,:)^T]. \tag{10}$$

Then, $\mathcal{L}_{(3)}(:,p)$ and $\mathcal{L}(i_p, j_p, :)$ are utilized to denote the pth pixel of $\mathcal{L}$, where $p = (j_p - 1)n_1 + i_p$. As introduced above, the neighbors of the pth pixel are defined as the pixels located in the square window centralized at the pth pixel with radius r, and these $(2r+1)^2 - 1$ neighbor pixels comprise the neighbor set NB_p, as shown in Fig. 7.

To constrict the value of the pth pixel to be close to that of its neighbors, the weight $W(p,q)$ between the pth and qth pixels is defined as

$$W(p,q) = \begin{cases} \exp\left(-\dfrac{\sqrt{(i_p - i_q)^2 + (j_p - j_q)^2}}{\sigma}\right), & q \in NB_p \\ 0, & q \notin NB_p. \end{cases} \tag{11}$$

Finally, based on the weight matrix W, the position-based Laplacian graph regularization is established as

$$\begin{aligned} \min_{\mathcal{L}_{(3)}} \ & \frac{1}{2} \sum_{p=1}^{n_1 n_2} \sum_{q=1}^{n_1 n_2} \|\mathcal{L}_{(3)}(:,p) - \mathcal{L}_{(3)}(:,q)\|_F^2 W(p,q) \\ &= \min_{\mathcal{L}_{(3)}} \ \text{Tr}\left(\mathcal{L}_{(3)} D \mathcal{L}_{(3)}^T\right) - \text{Tr}\left(\mathcal{L}_{(3)} W \mathcal{L}_{(3)}^T\right) \\ &= \min_{\mathcal{L}_{(3)}} \ \text{Tr}\left(\mathcal{L}_{(3)} G \mathcal{L}_{(3)}^T\right) \end{aligned} \tag{12}$$

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where Tr() denotes the trace of a matrix. D is a diagonal matrix whose diagonal element is $D(p, p) = \sum_{q=1}^{n_1 n_2} W(p, q)$. $G = D - W$ is the Laplacian matrix.

C. LPGTRPCA

To separate a tensor into a low-rank part and a noise part, the objective function of LPGTRPCA is developed as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}} \ \|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,2,1} + \beta \mathrm{Tr}\left(\mathcal{L}_{(3)} G \mathcal{L}_{(3)}^T\right) \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E} \end{aligned} \tag{13}$$

where $\mathcal{X}$, $\mathcal{L}$, and $\mathcal{E}$ are the original corrupted tensor, the low-rank part, and the noise part, respectively. α and β are two nonnegative tradeoff parameters.

The first term of the objective function ensures the low-rank property of $\mathcal{L}$. The second term requires the sparsity for each frontal slice of $\mathcal{E}$. The third term preserves the locality of the low-rank part $\mathcal{L}$ to relieve the impact of the residual noise. Optimizing this objective function decomposes a corrupted tensor into a low-rank part with preserved locality and a noise part with frontal slice sparsity.

It is noteworthy that the $l_{2,2,1}$ norm and the position-based locality preserving graph cooperate well in LPGTRPCA, rather than working separately. Specifically, both the $l_{2,2,1}$ norm and the position-based graph work in the frontal direction and ensure the smoothness of locality to relieve the impact of noise. Thus, LPGTRPCA is able to separate the low-rank part with little noise from a raw hyperspectral image and achieves robust classification results.

D. Connections to and Differences From Other Tensor-Based RPCA Methods

Like other methods, LPGTRPCA aims to partition a corrupted tensor $\mathcal{X}$ into a low-rank clean part $\mathcal{L}$ and a sparse noise part $\mathcal{E}$, but there are some differences. Table I lists the details of related tensor-based RPCA methods. We can see that differences occur in the low-rank constraint, the sparse property, and the regularization term. For the low-rank constraint, LPGTRPCA uses a newly defined nuclear norm [38]. Regarding the sparse property, the l_1 norm, the $l_{1,1,2}$ norm, and the $l_{2,1,2}$ norm were used in previous methods. TRPCA14 [32] uses the $l_{1,1,2}$ norm to discover the tubewise sparse noise of multilinear images, but the $l_{1,1,2}$ norm is not applicable for the noise distributed in hyperspectral images. The TRPCA16 [35], GTRPCA [47], IRTPCA [37], and TRPCA19 [38] utilize the l_1 norm to require the global sparsity of a tensor but obtain poor performance on hyperspectral image classification [39]. The ORTPCA [36] and LSSTRPCA [39] use the $l_{2,1,2}$ norm to extract the sparse noise of each lateral slice, but the $l_{2,1,2}$ norm may lead to uneven locality and, thus, limit the further improvement of the classification performance. Different from previous methods, LPGTRPCA uses the $l_{2,2,1}$ norm to extract the sparse noise from the frontal direction and ensure the smoothness of local areas. Finally, GTRPCA [47] and LPGTRPCA have an additional regularization term. For LPGTRPCA, the position-based Laplacian graph regularization term is added to relieve the impact of the residual noise remaining in the low-rank part. The position-based graph is more robust to noise than the intensity-based graph used in GTRPCA [47].

E. Optimization

The ADMM is employed to optimize the objective function of the LPGTRPCA. First, an auxiliary variable $\mathcal{Z}$ is introduced to modify objective function as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{Z},\mathcal{E}} \ \|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,2,1} + \beta \mathrm{Tr}\left(\mathcal{Z}_{(3)} G \mathcal{Z}_{(3)}^T\right) \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E}, \quad \mathcal{Z} = \mathcal{L}. \end{aligned} \tag{14}$$

Change the modified objective function into the augmented Lagrange formulation as

$$\begin{aligned} &F(\mathcal{L}, \mathcal{E}, \mathcal{Z}, \mathcal{Y}_1, \mathcal{Y}_2, \mu) \\ &\quad = \|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,2,1} + \beta \mathrm{Tr}\left(\mathcal{Z}_{(3)} G \mathcal{Z}_{(3)}^T\right) \\ &\qquad + \langle \mathcal{Y}_1, \mathcal{X} - \mathcal{L} - \mathcal{E}\rangle + \frac{\mu}{2}\|\mathcal{X} - \mathcal{L} - \mathcal{E}\|_F^2 \\ &\qquad + \langle \mathcal{Y}_2, \mathcal{Z} - \mathcal{L}\rangle + \frac{\mu}{2}\|\mathcal{Z} - \mathcal{L}\|_F^2 \end{aligned} \tag{15}$$

where $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are the tensors of Lagrange multipliers and μ is a nonnegative scalar.

The augmented Lagrange form is rewritten as

$$\begin{aligned} F(\mathcal{L}, \mathcal{E}, \mathcal{Z}, \mathcal{Y}_1, \mathcal{Y}_2, \mu) = \ &\|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,2,1} + \beta \mathrm{Tr}\left(\mathcal{Z}_{(3)} G \mathcal{Z}_{(3)}^T\right) \\ &+ \frac{\mu}{2}\left\|\mathcal{L} + \mathcal{S} - \mathcal{X} - \frac{\mathcal{Y}_1}{\mu}\right\|_F^2 \\ &+ \frac{\mu}{2}\left\|\mathcal{L} - \mathcal{Z} - \frac{\mathcal{Y}_2}{\mu}\right\|_F^2. \end{aligned} \tag{16}$$

1) Update $\mathcal{L}$: If other variables are fixed, $\mathcal{L}$ is updated by solving the following subproblem:

$$\begin{aligned} \mathcal{L} &= \operatorname*{argmin}_{\mathcal{L}} \ F(\mathcal{L}) \\ &= \operatorname*{argmin}_{\mathcal{L}} \|\mathcal{L}\|_* + \frac{\mu}{2}\left\|\mathcal{L} + \mathcal{E} - \mathcal{X} - \frac{\mathcal{Y}_1}{\mu}\right\|_F^2 \\ &\quad + \frac{\mu}{2}\left\|\mathcal{L} - \mathcal{Z} - \frac{\mathcal{Y}_2}{\mu}\right\|_F^2. \end{aligned} \tag{17}$$

Then, the above equation is written as

$$\operatorname*{argmin}_{\mathcal{L}} \ \frac{2}{\mu}\|\mathcal{L}\|_* + \frac{1}{2}\left\|\mathcal{L} - \frac{\mu(\mathcal{X} - \mathcal{S} + \mathcal{Z}) + \mathcal{Y}_1 + \mathcal{Y}_2}{2\mu}\right\|_F^2. \tag{18}$$

The tensor-SVT method [38] is used to solve (18) as

$$\mathcal{L} = \Gamma_{\frac{2}{\mu}}\left(\frac{\mu(\mathcal{X} - \mathcal{S} + \mathcal{Z}) + \mathcal{Y}_1 + \mathcal{Y}_2}{2\mu}\right). \tag{19}$$

Finally, for the $t+1$ iteration, $\mathcal{L}^{t+1}$ is updated as

$$\mathcal{L}^{t+1} = \Gamma_{\frac{2}{\mu^t}}\left(\frac{\mu^t(\mathcal{X} - \mathcal{E}^t + \mathcal{Z}^t) + \mathcal{Y}_1^t + \mathcal{Y}_2^t}{2\mu^t}\right). \tag{20}$$

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TABLE I
COMPARISON BETWEEN VARIOUS TENSOR-BASED RPCA METHODS

Method	Model	Low-rank constraint	Sparse property	Other regularization	Optimization
TRPCA14 [32]	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_{1,1,2}$	Nuclear norm [31], [33]	Tensor sparsity	–	ADMM
TRPCA16 [35]	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_1$	Nuclear norm [35]	Tensor sparsity	–	ADMM
GTRPCA [47]	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_1+\beta\sum_{i=1}^{n_3} Tr\big(\mathcal{L}(:,:,i)\mathcal{G}(:,:,i)\mathcal{L}(:,:,i)^T\big)$	Nuclear norm [35]	Tensor sparsity	Intensity-based Laplacian graph	ADMM
ORTPCA [36]	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_{2,1,2}$	Nuclear norm [35]	Lateral slice sparsity	–	ADMM
IRTPCA [37]	$\min_{\mathcal{L},\mathcal{E}} rank_i(\mathcal{L})+\alpha\|\mathcal{E}\|_1$	Tubal rank and core matrix rank [37]	Tensor sparsity	–	ADMM
TRPCA19 [38]	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_1$	Nuclear norm [38]	Tensor sparsity	–	ADMM
LSSTRPCA [39]	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_{2,1,2}$	Nuclear norm [35], [38]	Lateral slice sparsity	–	IALM
LPGTRPCA	$\min_{\mathcal{L},\mathcal{E}} \|\mathcal{L}\|_*+\alpha\|\mathcal{E}\|_{2,2,1}+\beta Tr\big(\mathcal{L}_{(3)}G\mathcal{L}_{(3)}^T\big)$	Nuclear norm [38]	Frontal slice sparsity	Position-based Laplacian graph	ADMM

2) Update $\mathcal{Z}$: If other variables are fixed, $\mathcal{Z}$ is updated by solving the following subproblem:

$$\begin{aligned}\mathcal{Z} &= \underset{\mathcal{Z}}{\operatorname{argmin}}\ F(\mathcal{Z})\\ &= \underset{\mathcal{Z}}{\operatorname{argmin}}\ \beta\mathrm{Tr}\big(\mathcal{Z}_{(3)}G\mathcal{Z}_{(3)}^T\big)+\frac{\mu}{2}\left\|\mathcal{Z}_{(3)}+\frac{\mathcal{Y}_{2(3)}}{\mu}-\mathcal{L}_{(3)}\right\|_F^2.\end{aligned} \tag{21}$$

Setting the gradient with respect to $\mathcal{Z}_{(3)}$ to zero, we have

$$\beta\mathcal{Z}_{(3)}G+\beta\mathcal{Z}_{(3)}G^T+\mu\mathcal{Z}_{(3)}+\mu\mathcal{Y}_{2(3)}-\mu\mathcal{L}_{(3)}=0. \tag{22}$$

According to (22), we have

$$\mathcal{Z}_{(3)}=\frac{\mu\mathcal{L}_{(3)}-\mathcal{Y}_{2(3)}}{\beta G+\beta G^T+\mu I}. \tag{23}$$

For the $t+1$ iteration, $\mathcal{Z}_{(3)}^{t+1}$ is computed as

$$\mathcal{Z}_{(3)}^{t+1}=\frac{\mu^t\mathcal{L}_{(3)}^{t+1}-\mathcal{Y}_{2(3)}^t}{\beta G+\beta G^T+\mu^t I}. \tag{24}$$

Finally, $\mathcal{Z}^{t+1}$ is yielded as

$$\mathcal{Z}^{t+1}=\mathrm{fold}\left(\mathcal{Z}_{(3)}^{t+1}\right) \tag{25}$$

where fold() denotes the tensor *mode-3* folding [52] corresponding to the *mode-3* unfolding given in (10).

3) Update $\mathcal{E}$: If other variables are fixed, $\mathcal{E}$ is updated by solving the following subproblem:

$$\begin{aligned}\mathcal{E} &= \underset{\mathcal{E}}{\operatorname{argmin}}\ F(\mathcal{E})\\ &= \underset{\mathcal{E}}{\operatorname{argmin}}\ \frac{\alpha}{\mu}\|\mathcal{E}\|_{2,2,1}+\frac{1}{2}\left\|\mathcal{E}-\left(\mathcal{X}+\frac{\mathcal{Y}_1}{\mu}-\mathcal{L}\right)\right\|_F^2.\end{aligned} \tag{26}$$

Formulate the above equation as

$$\underset{\mathcal{E}}{\operatorname{argmin}}\ \frac{\alpha}{\mu}\|\mathcal{E}_{(3)}\|_{2,1}+\frac{1}{2}\left\|\mathcal{E}_{(3)}-\left(\mathcal{X}_{(3)}+\frac{\mathcal{Y}_{1(3)}}{\mu}-\mathcal{L}_{(3)}\right)\right\|_F^2. \tag{27}$$

Set the gradient with respect to $\mathcal{E}_{(3)}$ to zero; we have (28)

$$\frac{\alpha}{\mu}D\mathcal{E}_{(3)}+\mathcal{E}_{(3)}-\mathcal{X}_{(3)}-\frac{\mathcal{Y}_{1(3)}}{\mu}+\mathcal{L}_{(3)}=0 \tag{28}$$

Algorithm 2 LPGTRPCA

Input: Tensor data $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}$, locality preserving graph $G\in\mathbb{R}^{n_1n_2\times n_1n_2}$, regularization parameters α and β.
Initialize: $t=0$, $\mathcal{L}^t=0$, $\mathcal{Z}^t=0$, $\mathcal{E}^t=0$, $\mathcal{Y}_1^t=0$, $\mathcal{Y}_2^t=0$, $\mu^t=10^{-3}$, $\mu_{max}=10^{10}$, $\epsilon=10^{-3}$.
Repeat: $t=t+1$.
Update the low-rank part $\mathcal{L}^{t+1}$ as (20).
Update the auxiliary variable $\mathcal{Z}^{t+1}$ as (25).
Update the sparse noise part $\mathcal{E}^{t+1}$ as (31).
Update $\mathcal{Y}_1^{t+1}$, $\mathcal{Y}_2^{t+1}$, and μ^{t+1} as (32), (33), and (34).
Until: All following criteria are satisfied.
$\left\|\mathcal{X}-\mathcal{L}^{t+1}-\mathcal{E}^{t+1}\right\|_F^2\le\epsilon$, $\left\|\mathcal{L}^{t+1}-\mathcal{L}^t\right\|_F^2\le\epsilon$, $\left\|\mathcal{E}^{t+1}-\mathcal{E}^t\right\|_F^2\le\epsilon$, $\left\|\mathcal{L}^t-\mathcal{Z}^t\right\|_F^2\le\epsilon$.
Output: $\mathcal{L}^{t+1}$.

where D is a diagonal matrix associated with $\mathcal{E}_{(3)}$, and $D(i,i)=(1/(\sum_{j=1}^{n_1n_2}\mathcal{E}_{(3)}(i,j)^2)^{1/2})$. According to (28), we have

$$\mathcal{E}_{(3)}=\left(\frac{\alpha}{\mu}D+I\right)^{-1}\left(\mathcal{X}_{(3)}+\frac{\mathcal{Y}_{1(3)}}{\mu}-\mathcal{L}_{(3)}\right). \tag{29}$$

For the $t+1$ iteration, $\mathcal{E}_{(3)}^{t+1}$ is computed as

$$\mathcal{E}_{(3)}^{t+1}=\left(\frac{\alpha}{\mu^t}D^t+I\right)^{-1}\left(\mathcal{X}_{(3)}+\frac{\mathcal{Y}_{1(3)}^t}{\mu^t}-\mathcal{L}_{(3)}^{t+1}\right). \tag{30}$$

Finally, $\mathcal{E}^{t+1}$ is obtained as

$$\mathcal{E}^{t+1}=\mathrm{fold}\left(\mathcal{E}_{(3)}^{t+1}\right). \tag{31}$$

4) Update $\mathcal{Y}_1$, $\mathcal{Y}_2$, and μ: Finally, $\mathcal{Y}_1$, $\mathcal{Y}_2$, and μ are updated as follows:

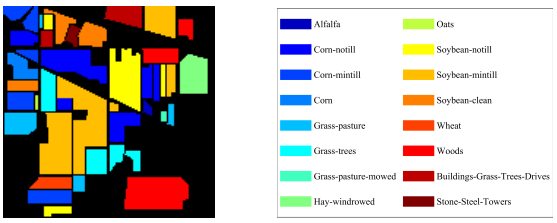
$$\mathcal{Y}_1^{t+1}=\mathcal{Y}_1^t+\mu^t(\mathcal{X}-\mathcal{L}^{t+1}-\mathcal{E}^{t+1}) \tag{32}$$

$$\mathcal{Y}_2^{t+1}=\mathcal{Y}_2^t+\mu^t(\mathcal{Z}^{t+1}-\mathcal{L}^{t+1}) \tag{33}$$

$$\mu^{t+1}=\min(1.1\mu^t,\mu_{\max}). \tag{34}$$

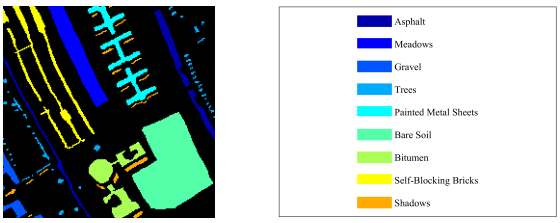
The proposed LPGTRPCA is summarized in Algorithm 2. After initialization, the update is repeated until convergence. Then, classification can be conducted on the low-rank part $\mathcal{L}$.

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(a) (b)

Fig. 8. (a) Ground truth and (b) land cover types of Indian Pines.



(a) (b)

Fig. 9. (a) Ground truth and (b) land cover types of Pavia University.

IV. Experiments

In this section, experiments were implemented on three real-world hyperspectral images to test the performance of LPGTRPCA. First, the classification results of LPGTRPCA were compared to that of eight classic and state-of-the-art RPCA methods to show efficiency. Second, an ablation study was carried out to prove the effect of the $l_{2,2,1}$ norm and the position-based Laplacian graph. Third, the tuning of the regularization parameter α and β was provided. Next, the influence of the radius r of the position-based graph was analyzed. Later, the convergence of LPGTRPCA was demonstrated. Finally, discussions were conducted to demonstrate the reliability and innovation of the proposed method.

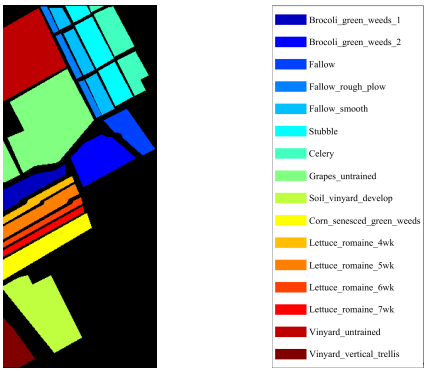
A. Datasets, Classifiers, and Performance Metrics

To show the superiority of LPGTROCA, three real-world hyperspectral image datasets were used in the experiments. The details of these datasets are introduced as follows.

1) Indian Pines [51]: The Indian Pines hyperspectral image was acquired by NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over West Lafayette, IN, USA, on June 12, 1992, which was taken from the Multi-Spectral Image Data Analysis System Group, Purdue University (https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html). This image contains 145 $\times$ 145 pixels of 16 land cover types. After band removal, 200 bands were used in the experiments. The ground truth was shown in Fig. 8.

2) Pavia University [53]: The Pavia University dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in Italy and was taken from the Computational Intelligence Group, Basque University (http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remot-e_Sensing_Scenes). This image consists of 610 $\times$ 340 pixels of nine land cover types, and 103 bands remained after band removal. A square part (from 156 to 375 rows and 21 to 240 columns) suffering from noise heavily was used for testing. The ground truth was shown in Fig. 9.

3) Salinas [53]: The Salinas dataset was acquired by the AVIRIS sensor over Salinas Valley, CA, USA, which was taken from the Computational Intelligence Group, Basque University (http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes). This dataset contains 512 $\times$ 217 pixels of 16 land cover types. After band removal, 204 bands were used in the experiments. The ground truth of this image was shown in Fig. 10.



(a) (b)

Fig. 10. (a) Ground truth and (b) land cover types of Salinas.

TABLE II
Number of the Samples of Training Data and Testing Data of Indian Pines

Class	Name	Training Data	Testing Data
1	Alfalfa	4	42
2	Corn-notill	142	1286
3	Corn-mintill	83	747
4	Corn	23	214
5	Grass-pasture	48	435
6	Grass-trees	73	657
7	Grass-pasture-mowed	2	26
8	Hay-windrowed	47	431
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	245	2210
12	Soybean-clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Buildings-Grass-Trees-Drives	38	348
16	Stone-Steel-Towers	9	84

The support vector machine (SVM) with radial basis function was used as the classifier, and the parameters were set via cross-validation. For all three datasets, 10% of pixels of each class were randomly selected to train SVM, and the remaining 90% were used as testing data. The numbers of pixels of training and testing data were shown in Tables II–IV.

Three metrics, including the average class classification accuracy (AA) [54], [55], the overall classification accuracy (OA) [54], [55], and the kappa coefficient (KC) [54], [55], were used to measure the classification performance. All tested methods were conducted ten times, and the average results were provided.

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TABLE III
NUMBER OF THE SAMPLES OF TRAINING DATA AND TESTING DATA OF PAVIA UNIVERSITY

Class	Name	Training Data	Testing Data
1	Asphalt	129	1161
2	Meadows	174	1570
3	Gravel	47	419
4	Trees	51	455
5	Painted Metal Sheets	129	1161
6	Bare Soil	503	4552
7	Bitumen	132	1195
8	Self-Blocking Bricks	156	1410
9	Shadows	46	406

TABLE IV
NUMBER OF THE SAMPLES OF TRAINING DATA AND TESTING DATA OF SALINAS

Class	Name	Training Data	Testing Data
1	Brocoli_green_weeds_1	200	1809
2	Brocoli_green_weeds_2	372	3354
3	Fallow	197	1779
4	Fallow_rough_plow	139	1255
5	Fallow_smooth	267	2411
6	Stubble	395	3564
7	Celery	357	3222
8	Grapes_untrained	1127	10144
9	Soil_vinyard_develop	620	5583
10	Corn_senesced_green_weeds	327	2951
11	Lettuce_romaine_4wk	106	962
12	Lettuce_romaine_5wk	192	1735
13	Lettuce_romaine_6wk	91	825
14	Lettuce_romaine_7wk	107	963
15	Vinyard_untrained	726	6542
16	Vinyard_vertical_trellis	180	1627

B. Classification Performance

The classification performance of LPGTRPCA was compared to that of classic and state-of-the-art methods, including RPCA [21], CWRPCA [22], TRPCA16 [35], GTRPCA [47], ORTPCA [36], IRTPCA [37], TRPCA19 [38], and LSSTRPCA [39]. The parameter settings of these methods were based on related work [36], [37], [39], [47]. On the Indian Pines dataset, the regularization parameters of RPCA, CWRPCA, TRPCA16, ORTPCA, TRPCA19, and LSSTRPCA were set to 0.1, 0.1, 0.001, 0.5, 0.001, and 0.5, respectively. The two parameters of IRTPCA were set to $(1/(\max(n1, n2, n3))^{1/2})$ and $(2/(\max(n1, n2)n3)^{1/2})$. The parameters of GTRPCA were set to 1 and 3, and the radius of the graph was set to 1. For LPGTRPCA, α and β were set to 1 and 3, respectively, and the radius r of the locality preserving graph was set to 1.

On the Pavia University dataset, the regularization parameters of RPCA, CWRPCA, TRPCA16, ORTPCA, TRPCA19, and LSSTRPCA were set to 0.1, 0.1, 0.001, 0.5, 0.001, and 0.2, respectively. The two parameters of IRTPCA were set to $(1/(\max(n1, n2, n3))^{1/2})$ and $(2/(\max(n1, n2)n3)^{1/2})$. The parameters of GTRPCA were set to 1 and 3, and the radius of the graph was set to 1. For LPGTRPCA, α and β were set to 3 and 3, respectively, and the radius r of the locality preserving graph was set to 1.

On the Salinas dataset, the parameters of RPCA, CWRPCA, TRPCA16, ORTPCA, TRPCA19, and LSSTRPCA were set to 0.05, 0.05, 0.003, 0.2, 0.003, and 0.2, respectively. The parameters of IRTPCA were set as $(1/(\max(n1, n2, n3))^{1/2})$ and $(2/(\max(n1, n2)n3)^{1/2})$. The two parameters of GTRPCA were set to 0.2 and 0,01, and the radius of the graph was set to 1. For LPGTRPCA, α and β were set to 0.2 and 0.01, respectively, and the radius r of the locality preserving graph was set to 1.

Table V lists the classification performance of all tested methods on the Indian Pines dataset. We can see that all tensor-based methods except GTRPCA outperformed better than matrix-based methods. This shows the advantages of tensor-based methods at discovering inherent structures across spectral bands. ORTPCA, LSSTRPCA, and LPGTRPCA outperformed the other tensor-based methods because the $l_{2,1,2}$ norm and the $l_{2,2,1}$ norm can efficiently handle the sparse noise in some spectral bands and local areas. The most important observation from Table V is that LPGTRPCA significantly improved classification results compared to other tested methods. LPGTRPCA obtained AA, OA, and KC values of 96.21%, 98.22%, and 97.97%, respectively, with second-best results of LSSTRPCA of 92.03%, 93.66%, and 92.77%. The huge improvement of classification results demonstrates the superiority of LPGTRPCA. Table V shows that LPGTRPCA obtained the highest accuracy on most single classes (13 of 16). In some classes (such as Corn-notill, Corn, Soybean-notill, Soybean-mintill, and Stone-Steal-Towers), our method's accuracy was around 6% better than the best obtained by other methods. It is noteworthy that another graph-based method, GTRPCA, obtained poor classification results (75.49%, 80.86%, and 78.11% in AA, OA, and KC, respectively) because the l_1 norm that it uses cannot sufficiently extract the sparse noise in hyperspectral images, and the intensity-based graph is sensitive to the noise remaining in the low-rank image. In contrast, the $l_{2,2,1}$ norm and the position-based graph used by LPGTRPCA are more suitable for hyperspectral image processing. Fig. 11 shows the classification maps of all methods on Indian Pines. From Fig. 11(j), we can see that just a few pixels were classified incorrectly by LPGTRPCA, while other methods suffered greatly from sparse noise, as shown in Fig. 11(b)–(i). Based on these observations, LPGTRPCA performed best among the tested methods on the Indian Pines dataset.

Similar results were observed on the Pavia University and Salinas datasets, as shown in Tables VI and VII. In Pavia University, LPGTRPCA obtained the highest accuracy on seven single classes and significantly promoted AA, OA, and KC. For example, the OA of LPGTRPCA was 99.52%, while the second best one achieved by LSSTRPCA was 97.02%. In Salinas, LPGTRPCA also performed best on eight single classes and got the highest AA, OA, and KC. From Figs. 12 and 13, it can be seen that the misclassification only appeared on a very small number of pixels in the classification result of LPGTRPCA, while the classification maps of other methods still suffered from sparse noise. Thus, we can say that LPGTRPCA defeated all other tested methods on Pavia University and Salinas.

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TABLE V
CLASSIFICATION PERFORMANCE OF TESTED METHODS ON INDIAN PINES

Class	Name	RPCA	CWRPCA	TRPCA16	GTRPCA	ORTPCA	IRTPCA	TRPCA19	LSSTRPCA	LPGTRPCA
1	Alfalfa	56.10±9.13	62.85±14.20	83.71±7.85	50.00±19.21	84.48±10.95	***89.51±6.71**	76.77±12.46	88.37±9.71	87.58±16.92
2	Corn-notill	78.35±1.93	80.75±2.49	89.97±1.41	78.16±2.50	92.57±1.54	86.23±1.73	89.09±2.21	91.68±1.96	***98.35±0.98**
3	Corn-mintill	68.74±4.72	69.61±2.62	92.81±2.67	66.89±3.95	91.85±1.93	87.93±1.56	93.03±1.70	92.49±2.66	***97.60±1.78**
4	Corn	56.86±8.27	61.40±8.61	90.43±5.01	60.73±5.35	87.95±6.27	85.42±5.06	89.27±6.88	86.42±5.56	***96.35±2.87**
5	Grass-pasture	89.15±3.59	90.28±2.18	91.55±2.69	88.02±3.83	95.68±1.68	93.31±1.20	91.23±1.89	94.92±2.37	***95.78±3.00**
6	Grass-trees	94.83±1.93	96.44±1.88	98.71±0.70	96.00±1.68	99.12±0.70	97.54±2.14	98.38±1.29	***99.48±0.58**	99.14±0.98
7	Grass-pasture-mowed	78.00±11.35	76.12±16.92	83.75±9.64	79.20±5.59	86.34±16.39	***95.60±3.50**	82.80±12.80	92.15±5.69	93.42±12.95
8	Hay-windrowed	97.04±2.27	98.78±1.21	98.78±1.85	97.48±2.21	98.68±1.24	99.07±0.75	98.83±1.16	99.31±0.39	***99.72±0.46**
9	Oats	34.44±20.75	36.00±13.11	74.44±23.16	31.55±14.68	79.44±19.95	64.82±24.97	77.05±18.01	71.67±24.91	***81.23±21.61**
10	Soybean-notill	68.68±4.90	68.67±3.10	85.96±3.15	73.27±3.69	88.03±2.46	75.76±3.48	86.89±3.09	88.20±1.65	***97.15±1.44**
11	Soybean-mintill	74.73±1.87	75.94±2.45	90.92±1.35	81.87±2.24	92.49±1.84	84.00±1.76	90.75±1.52	92.83±1.63	***98.61±0.60**
12	Soybean-clean	69.21±4.07	73.00±3.43	90.31±2.29	70.88±3.33	92.37±2.60	86.88±2.62	90.95±1.59	92.84±2.05	***96.76±1.43**
13	Wheat	97.19±2.18	97.59±3.25	98.10±1.15	96.54±2.87	98.16±1.03	98.49±2.19	98.43±1.13	98.97±0.97	***99.68±0.45**
14	Woods	92.63±2.06	93.42±2.20	98.88±0.46	94.42±1.10	99.24±0.38	96.06±1.02	98.91±0.49	98.38±1.20	***99.68±0.30**
15	Buildings-Grass-Trees-Drives	58.72±4.74	57.78±5.06	96.74±1.90	54.82±3.06	89.96±4.16	77.86±4.03	96.80±1.12	92.47±2.47	***98.82±1.90**
16	Stone-Steel-Towers	90.27±5.12	90.72±4.12	90.69±7.16	87.95±4.91	93.05±3.91	90.11±5.24	91.34±7.06	92.30±5.00	***99.52±0.84**
	AA(%)	75.31±1.73	76.83±1.32	90.99±1.56	75.49±1.34	91.84±1.21	88.04±1.78	90.66±1.87	92.03±1.88	***96.21±1.71**
	OA(%)	78.64±0.47	80.01±0.80	92.64±0.28	80.86±0.49	93.60±0.40	87.77±0.33	92.54±0.56	93.66±0.56	***98.22±0.44**
	KC(%)	75.64±0.55	77.07±0.91	91.61±0.33	78.11±0.55	92.71±0.45	86.07±0.37	91.50±0.64	92.77±0.64	***97.97±0.50**

*** and bold fonts denote the best among the group.**

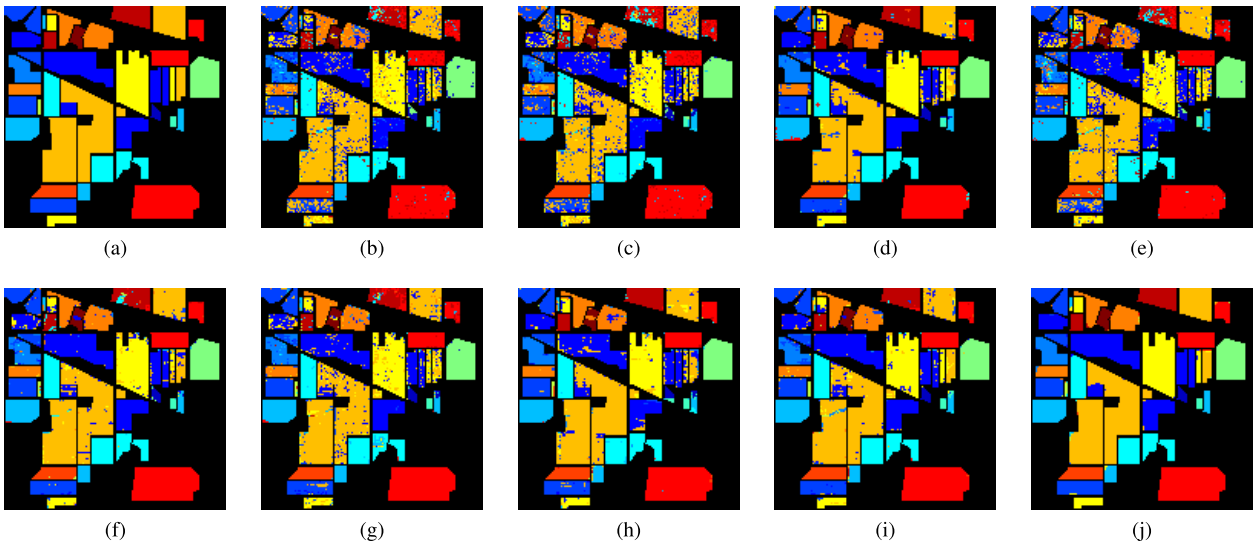


Fig. 11. Classification maps of tested methods on Indian Pines. (a) Ground truth. (b) RPCA. (c) CWRPCA. (d) TRPCA16. (e) GTRPCA. (f) ORTPCA. (g) IRTPCA. (h) TRPCA19. (i) LSSTRPCA. (j) LPGTRPCA.

Comparing Tables V–VII, we can find that the promotion achieved by the proposed LPGTRPCA was significant on Indian Pines and Pavia University, but not remarkable on Salinas. Specifically, compared with the second best result obtained by LSSTRPCA, the promotion of the OA was 4.56% and 2.50% on Indian Pines and Pavia University but only 0.08% on Salinas. We think the reason was that the current methods, such as LSSTRPCA, could get good performance in Salinas, as the noise in this dataset was not very heavy. Thus, the promotion of our method on Salinas was not significant. In contrast, the Indian Pines and Pavia University datasets contained more noise than Salinas, and the performance of current methods deteriorated obviously. The proposed LPGTRPCA could efficiently separate the low-rank part with little noise from noisy hyperspectral images for guaranteeing good classification results, so the promotion was remarkable on these two datasets. This interesting observation reflected the ability of LPGTRPCA to deal with hyperspectral images containing heavy noise.

C. Ablation Study

An ablation study was conducted in this part to show the effectiveness of the tensor $l_{2,2,1}$ norm and the position-based graph. Four methods, including LSSTRPCA, M1, M2, and

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TABLE VI
CLASSIFICATION PERFORMANCE OF TESTED METHODS ON PAVIA UNIVERSITY

Class	Name	RPCA	CWRPCA	TRPCA16	GTRPCA	ORTPCA	IRTPCA	TRPCA19	LSSTRPCA	LPGTRPCA
1	Asphalt	87.44±2.23	88.57±2.62	96.12±1.41	86.82±1.59	93.95±1.14	91.35±2.94	96.12±1.40	92.82±1.65	***99.18±0.50**
2	Meadows	95.67±1.46	96.02±0.98	98.75±0.49	95.92±1.07	98.85±0.60	97.64±1.25	98.52±0.89	99.28±0.33	***100.00±0.00**
3	Gravel	76.87±5.04	76.15±8.32	82.21±4.95	75.85±5.45	79.64±4.80	83.80±3.01	84.63±6.30	79.79±5.25	***96.15±2.05**
4	Trees	97.82±0.94	96.64±1.40	94.16±1.94	97.58±1.41	97.51±1.63	98.17±0.63	93.27±2.31	***98.66±0.65**	98.55±1.58
5	Painted Metal Sheets	99.45±0.26	99.58±0.21	99.72±0.12	99.41±0.29	99.74±0.13	99.38±0.40	99.82±0.16	99.81±0.16	***100.00±0.00**
6	Bare Soil	97.87±0.57	97.98±0.48	98.66±0.49	97.63±0.70	99.57±0.30	98.68±0.36	98.67±0.46	99.51±0.31	***99.99±0.02**
7	Bitumen	89.63±2.72	90.18±2.56	97.20±1.38	88.53±2.96	95.07±1.09	92.50±1.18	97.21±1.27	95.39±1.16	***99.58±0.35**
8	Self-Blocking Bricks	94.08±1.42	94.00±1.83	95.15±2.12	93.72±1.62	94.85±1.28	94.63±1.59	94.80±2.36	92.83±2.95	***99.46±0.32**
9	Shadows	99.93±0.12	***99.98±0.08**	99.43±0.49	99.90±0.17	99.75±0.31	99.90±0.13	98.53±0.72	99.68±0.29	96.76±2.03
	AA(%)	93.20±0.61	93.23±0.99	95.71±0.57	92.82±0.56	95.44±0.68	95.12±0.39	95.73±0.94	95.31±0.76	***98.85±0.31**
	OA(%)	94.87±0.39	95.06±0.39	97.29±0.38	94.56±0.38	97.24±0.27	96.37±0.36	97.26±0.42	97.02±0.45	***99.52±0.11**
	KC(%)	93.62±0.48	93.85±0.49	96.63±0.48	93.23±0.47	96.57±0.34	95.49±0.44	96.59±0.53	96.29±0.56	***99.41±0.14**

*** and bold fonts denote the best among the group.**

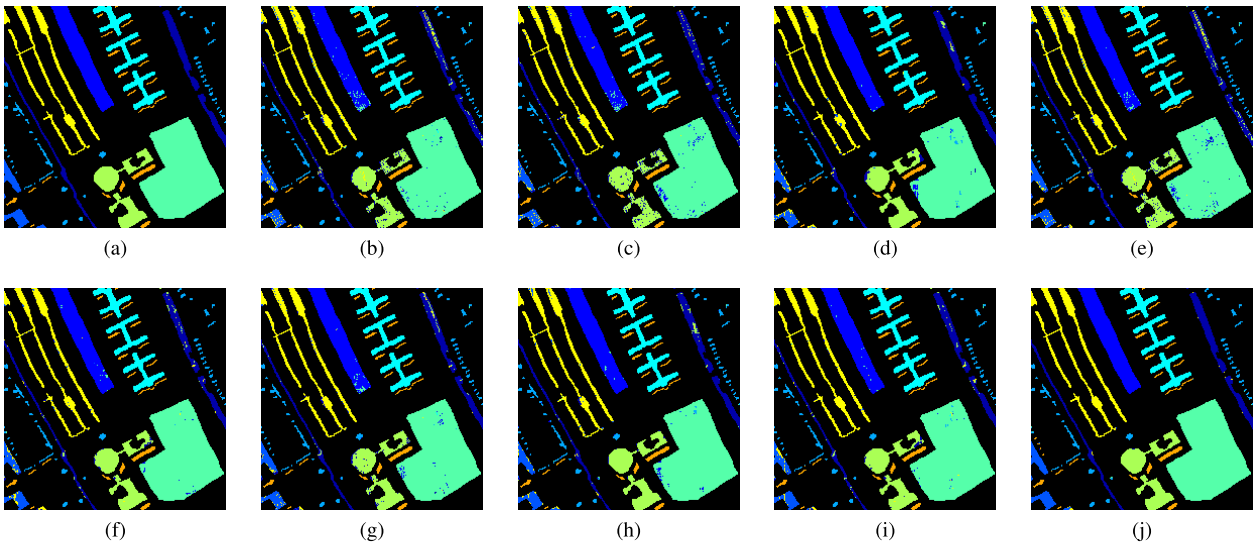


Fig. 12. Classification maps of tested methods on Pavia University. (a) Ground truth. (b) RPCA. (c) CWRPCA. (d) TRPCA16. (e) GTRPCA. (f) ORTPCA. (g) IRTPCA. (h) TRPCA19. (i) LSSTRPCA. (j) LPGTRPCA.

LPGTRPCA, were compared. The objective functions of M1 and M2 were given in (35) and (36). Specifically, LSSTRPCA and M2 used the tensor $l_{2,1,2}$ norm to require the lateral slice sparsity of the noise part; meanwhile, M1 and LPGTRPCA utilized the newly defined tensor $l_{2,2,1}$ norm for frontal slice sparsity. In addition, M2 and LPGTRPCA employed the position-based Laplacian graph to further lessen the impact of the residual noise. On three datasets, the parameters of M1 and M2 were the same as the ones of LSSTRPCA and LPGTRPCA, which were provided in the classification performance part. The classification results were listed in Table VIII

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}} \ \|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,2,1} \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E} \end{aligned} \tag{35}$$

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}} \ \|\mathcal{L}\|_* + \alpha\|\mathcal{E}\|_{2,1,2} + \beta \mathrm{Tr}\left(\mathcal{L}_{(3)} G \mathcal{L}_{(3)}^T\right) \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E}. \end{aligned} \tag{36}$$

From Table VIII, we had three observations. First, LSSTRPCA and M1 obtained similar results on all three datasets. This means that the proposed tensor $l_{2,2,1}$ norm can extract the noise of hyperspectral images as efficiently as the tensor $l_{2,1,2}$ norm. Second, M2 and LPGTRPCA performed better than LSSTRPCA and M1. For example, the OA of M2 and the LPGTRPCA on Indian Pines were 97.79% and 98.22%; meanwhile, the ones of LSSTRPCA and M1 on Indian Pines were only 93.66% and 93.86%. The remarkable promotion proved the effectiveness of the position-based graph on relieving the residual noise by preserving local structures. As shown in Figs. 14 and 15, there were few sparse errors in the classification maps of M2 and LPGTRPCA. Third, LPGTRPCA defeated M2, especially on Indian Pines and Pavia University. This reflected that the $l_{2,2,1}$ norm outperformed the $l_{2,1,2}$ norm when cooperated with the position-based Laplacian graph. The deficiency of the tensor

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TABLE VII
CLASSIFICATION PERFORMANCE OF TESTED METHODS ON SALINAS

Class	Name	RPCA	CWRPCA	TRPCA16	GTRPCA	ORTPCA	IRTPCA	TRPCA19	LSSTRPCA	LPGTRPCA
1	Brocoli_green_weeds_1	99.30±0.31	99.70±0.22	99.65±0.27	99.64±0.26	***99.83±0.18**	99.73±0.28	99.65±0.30	99.60±0.61	99.65±0.23
2	Brocoli_green_weeds_2	99.77±0.10	99.78±0.21	99.91±0.15	99.87±0.10	99.85±0.10	99.91±0.08	99.91±0.11	99.73±0.19	***99.95±0.10**
3	Fallow	99.28±0.41	99.67±0.15	99.87±0.10	99.73±0.25	99.84±0.11	99.86±0.11	99.83±0.12	99.71±0.17	***99.99±0.04**
4	Fallow_rough_plow	99.18±0.50	99.25±0.32	***99.37±0.39**	99.29±0.37	98.84±0.36	99.32±0.29	99.25±0.35	98.65±0.59	98.82±0.40
5	Fallow_smooth	98.93±0.34	99.10±0.32	***99.53±0.26**	99.17±0.22	99.01±0.43	99.29±0.45	99.44±0.21	99.16±0.51	99.19±0.37
6	Stubble	99.83±0.10	99.84±0.07	99.87±0.08	99.84±0.08	99.92±0.06	99.89±0.07	99.89±0.07	***99.95±0.02**	99.89±0.09
7	Celery	99.57±0.16	99.61±0.20	99.65±0.18	99.64±0.14	99.75±0.16	99.74±0.19	99.78±0.13	99.80±0.13	***99.84±0.15**
8	Grapes_untrained	87.39±0.57	88.63±0.87	96.41±0.36	89.39±0.57	99.53±0.19	90.67±0.77	95.99±0.45	99.57±0.09	***99.72±0.06**
9	Soil_vinyard_develop	99.87±0.07	99.83±0.11	99.86±0.07	99.86±0.10	99.91±0.09	99.92±0.06	99.87±0.15	99.92±0.06	***99.98±0.02**
10	Corn_senesced_green_weeds	92.41±0.57	97.30±0.55	98.32±0.51	97.17±0.46	98.88±0.45	98.18±0.53	98.23±0.51	99.06±0.31	***99.36±0.26**
11	Lettuce_romaine_4wk	96.05±1.57	98.98±0.75	98.82±0.51	98.06±1.04	98.45±0.69	99.30±0.52	***99.41±0.30**	99.05±0.26	98.83±0.65
12	Lettuce_romaine_5wk	99.71±0.16	99.89±0.13	99.95±0.11	99.91±0.09	99.95±0.07	99.90±0.10	99.95±0.06	***99.99±0.02**	99.32±0.56
13	Lettuce_romaine_6wk	97.75±1.08	98.40±0.89	99.15±0.75	98.70±0.87	99.12±0.74	***99.24±0.66**	99.06±0.91	99.06±0.87	98.66±0.63
14	Lettuce_romaine_7wk	94.86±1.68	97.28±1.13	98.37±1.55	97.15±1.77	98.68±0.77	98.00±1.12	98.40±0.76	***98.88±0.80**	98.86±0.85
15	Vinyard_untrained	66.22±1.28	75.21±0.88	93.66±0.61	74.90±0.77	99.52±0.17	81.51±1.05	93.40±0.81	99.46±0.22	***99.62±0.23**
16	Vinyard_vertical_trellis	98.21±0.58	99.04±0.38	98.70±0.62	98.77±0.47	99.55±0.45	98.69±0.48	98.46±0.60	99.56±0.30	***99.69±0.51**
	AA(%)	95.52±0.15	96.97±0.12	98.82±0.14	96.94±0.18	99.41±0.09	97.70±0.09	98.78±0.10	99.45±0.08	***99.46±0.10**
	OA(%)	91.89±0.14	93.85±0.13	98.07±0.08	93.95±0.14	99.55±0.06	95.24±0.09	97.95±0.07	99.56±0.07	***99.64±0.07**
	KC(%)	90.96±0.15	93.14±0.14	97.85±0.09	93.25±0.15	99.50±0.07	94.70±0.10	97.72±0.08	99.51±0.07	***99.60±0.08**

*** and bold fonts denote the best among the group.**

TABLE VIII
CLASSIFICATION RESULTS OF SEVERAL METHODS FOR ABLATION STUDY

Indian Pines					Pavia University					Salinas				
	LSSTRPCA	M1	M2	LPGTRPCA		LSSTRPCA	M1	M2	LPGTRPCA		LSSTRPCA	M1	M2	LPGTRPCA
AA(%)	92.03	92.10	95.45	***96.21**	AA(%)	95.31	95.41	98.54	***98.85**	AA(%)	99.45	99.44	99.39	***99.46**
OA(%)	93.66	93.86	97.79	***98.22**	OA(%)	97.02	97.09	99.25	***99.52**	OA(%)	99.56	99.54	99.60	***99.64**
KC(%)	92.77	93.01	97.48	***97.97**	KC(%)	96.29	96.32	99.07	***99.41**	KC(%)	99.51	99.50	99.56	***99.60**

*** and bold fonts denote the best among the group.**

TABLE IX
OA (%) OF LPGTRPCA WITH DIFFERENT VALUES OF REGULARIZATION PARAMETERS α AND β ON INDIAN PINES

		β												
		0.001	**0.005**	**0.01**	**0.05**	**0.1**	**0.2**	**0.5**	**1**	**1.5**	**2**	**3**	**5**	**10**
α	**0.001**	33.60	34.24	34.54	37.72	41.07	44.79	43.59	45.43	45.88	45.65	45.62	44.80	44.82
	0.005	33.67	34.31	34.14	36.42	37.77	40.09	43.92	43.26	43.23	42.90	44.23	43.51	44.50
	0.01	52.21	56.62	60.56	67.32	69.99	73.77	73.66	78.73	79.15	79.20	79.66	81.19	87.33
	0.05	88.05	88.80	88.93	90.75	92.27	93.99	95.15	95.50	95.30	95.57	95.47	95.57	94.77
	0.1	92.40	93.16	93.59	94.55	95.00	95.70	96.07	96.14	96.46	96.26	96.49	96.79	96.63
	0.2	94.24	94.41	94.80	95.60	96.32	96.62	**97.46**	**97.33**	**97.43**	**97.16**	96.78	96.79	96.85
	0.5	93.97	94.16	94.55	95.72	96.76	**97.01**	**97.64**	**97.57**	**97.91**	**97.89**	**97.57**	**97.74**	**97.27**
	1	84.04	83.86	84.90	88.38	92.32	95.80	**97.47**	**98.10**	**98.08**	**98.12**	**98.22**	**97.85**	**97.74**
	1.5	80.02	80.37	80.01	82.54	84.70	90.39	96.62	**97.47**	**98.02**	**97.89**	**98.01**	**98.22**	**97.98**
	2	79.23	79.03	78.95	79.52	81.71	85.57	94.43	**97.05**	**97.48**	**97.61**	**97.91**	**98.21**	**98.14**
	3	78.59	78.35	78.69	78.28	79.44	80.97	89.13	95.43	96.53	**97.10**	**97.59**	**97.95**	**98.04**
	5	78.47	78.79	78.91	78.64	78.77	79.06	81.99	90.16	93.82	95.33	96.58	**97.66**	**97.98**
	10	78.58	78.72	78.48	78.63	78.37	78.44	78.64	81.46	85.19	89.03	93.00	95.96	**97.36**

bold underline fonts denote the overall classification higher than 97%.

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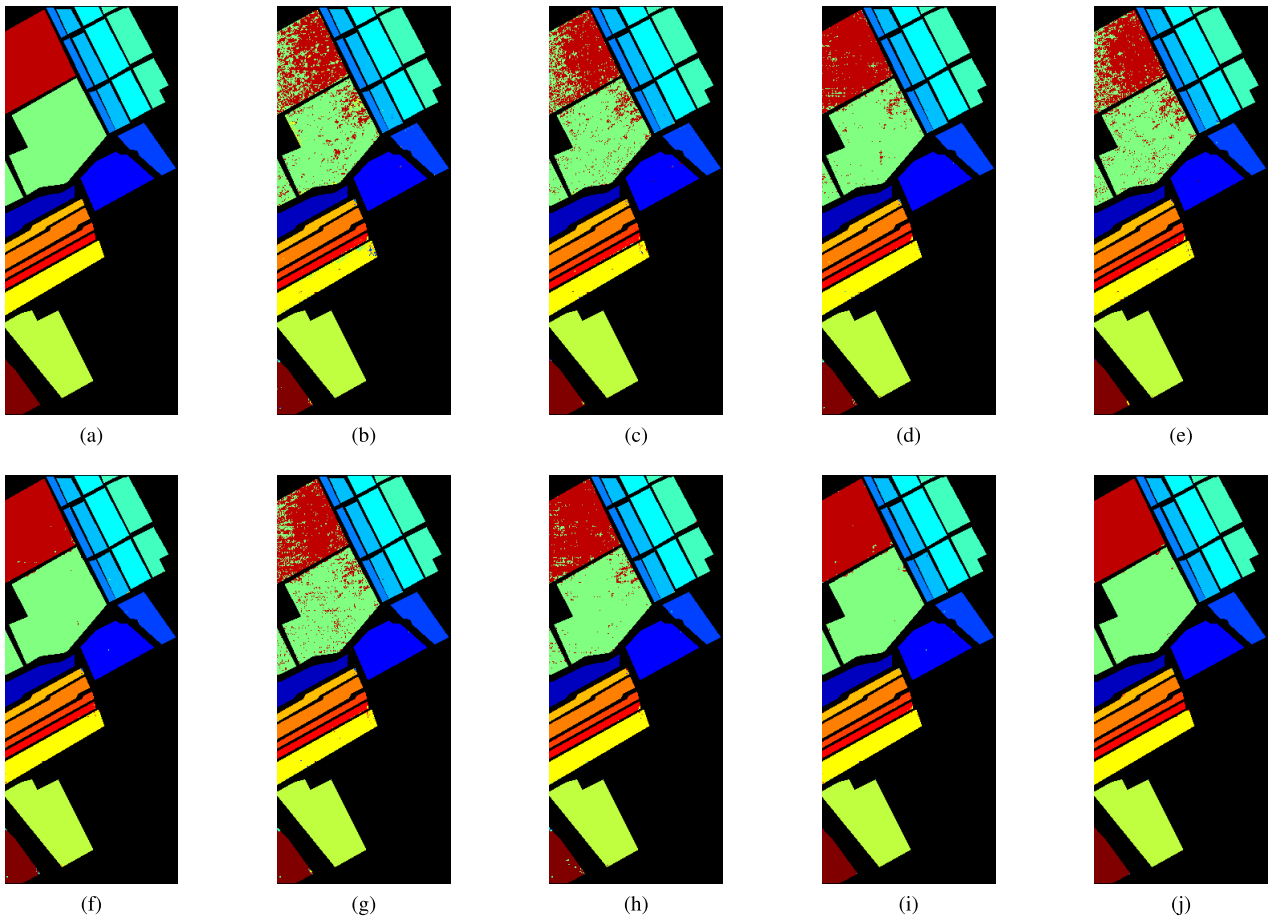


Fig. 13. Classification maps of tested methods on Salinas. (a) Ground truth. (b) RPCA. (c) CWRPCA. (d) TRPCA16. (e) GTRPCA. (f) ORTPCA. (g) IRTPCA. (h) TRPCA19. (i) LSSTRPCA. (j) LPGTRPCA.

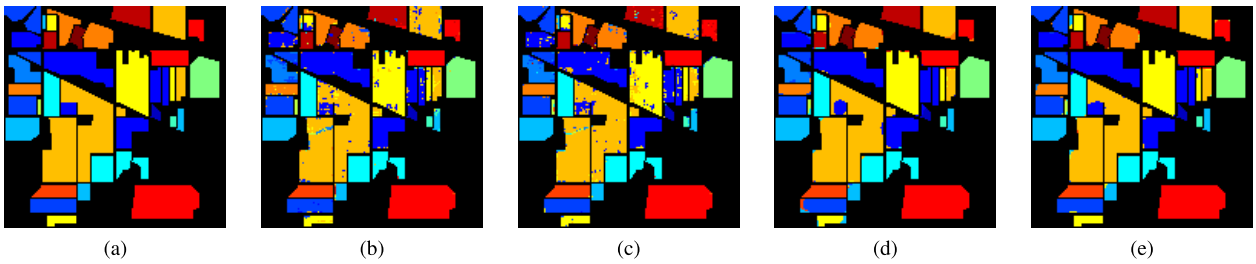


Fig. 14. Classification maps of LSSTRPCA, M1, M2, and LPGTRPCA on Indian Pines. (a) Ground truth. (b) LSSTRPCA. (c) M1. (d) M2. (e) LPGTRPCA.

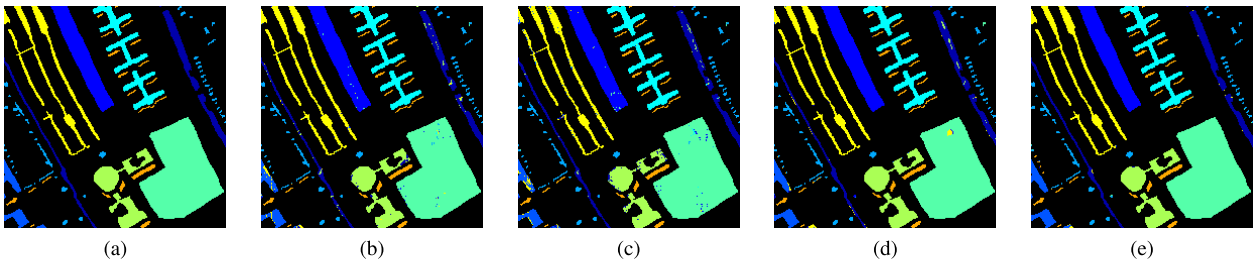
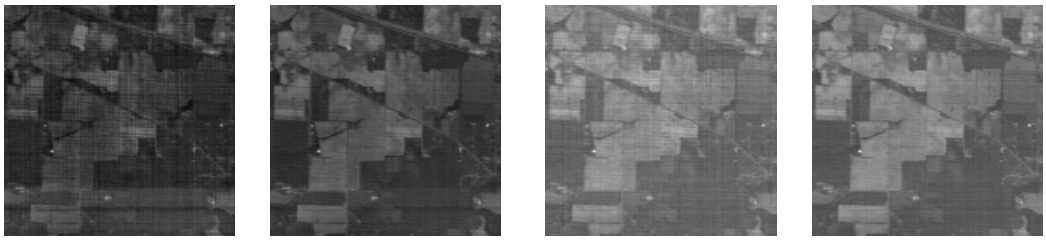


Fig. 15. Classification maps of LSSTRPCA, M1, M2, and LPGTRPCA on Pavia University. (a) Ground truth. (b) LSSTRPCA. (c) M1. (d) M2. (e) LPGTRPCA.

$l_{2,2,1}$ norm is that it resulted in the uneven locality, as shown in Fig. 16(a) and (c), and, thus, limited the effect of the position-based graph. In contrast, as shown in Fig. 16(b) and (d), the new defined $l_{2,2,1}$ norm guaranteed the smoothness of the locality. Thus, the $l_{2,2,1}$ norm cooperated better with the position-based Laplacian graph for good results.

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(a) (b) (c) (d)

Fig. 16. (a) and (b) First band and (c) and (d) last band of the low-rank images obtained by LSSTRPCA and M1 on Indian Pines.

TABLE X
OA (%) OF LPGTRPCA WITH DIFFERENT VALUES OF REGULARIZATION PARAMETERS α AND β ON PAVIA UNIVERSITY

		β												
		0.001	**0.005**	**0.01**	**0.05**	**0.1**	**0.2**	**0.5**	**1**	**1.5**	**2**	**3**	**5**	**10**
	0.001	36.92	36.91	36.90	36.90	36.90	36.88	36.92	36.89	36.92	36.89	36.91	36.92	36.90
	0.005	47.05	48.57	49.47	52.48	54.48	58.79	63.35	66.62	66.99	66.38	65.07	63.57	63.86
	0.01	46.85	48.13	49.11	52.07	53.18	56.14	60.40	63.29	66.70	67.07	66.53	65.22	63.94
	0.05	71.68	73.05	73.78	78.30	81.27	83.04	84.99	87.68	89.80	90.31	88.73	89.31	92.04
	0.1	85.61	86.53	86.82	88.31	89.00	91.11	93.01	93.53	93.92	94.02	94.01	94.02	93.77
	0.2	95.79	95.88	96.31	96.98	97.31	97.76	97.71	97.27	96.95	96.70	96.66	96.36	96.20
α	**0.5**	97.06	97.17	96.88	97.49	98.05	98.67	**99.24**	**99.08**	98.96	98.69	98.35	98.10	97.59
	1	96.82	96.85	96.82	97.22	97.71	98.33	**99.16**	**99.29**	**99.29**	**99.31**	**99.30**	98.85	98.68
	1.5	96.37	96.13	96.30	96.57	96.85	98.13	98.89	**99.32**	**99.47**	**99.45**	**99.35**	**99.15**	**99.07**
	2	96.02	96.31	96.15	96.19	96.28	97.39	98.70	**99.21**	**99.51**	**99.43**	**99.50**	**99.31**	**99.09**
	3	95.58	95.77	95.68	95.74	95.75	96.03	98.30	**99.06**	**99.18**	**99.35**	**99.52**	**99.41**	**99.21**
	5	95.28	95.31	95.09	95.36	95.32	95.30	95.72	98.33	98.92	98.98	**99.21**	**99.43**	**99.35**
	10	94.78	94.65	94.83	94.96	94.93	94.94	94.92	95.50	97.35	98.28	98.79	**99.20**	**99.40**

bold underline fonts denote the overall classification higher than 99%.

TABLE XI
OA (%) OF LPGTRPCA WITH DIFFERENT VALUES OF REGULARIZATION PARAMETERS α AND β ON SALINAS

		β												
		0.001	**0.005**	**0.01**	**0.05**	**0.1**	**0.2**	**0.5**	**1**	**1.5**	**2**	**3**	**5**	**10**
	0.001	29.24	27.87	27.51	27.32	27.32	27.31	27.15	27.02	26.76	26.91	27.21	29.25	31.21
	0.005	67.42	58.06	53.22	45.45	43.13	41.44	40.82	39.36	39.32	39.93	39.02	36.19	36.16
	0.01	93.42	89.73	86.35	70.19	63.51	59.33	50.85	43.09	40.77	40.27	39.53	39.30	39.82
	0.05	**99.28**	**99.28**	**99.29**	**99.23**	98.83	96.85	91.43	84.22	77.98	72.75	65.88	56.68	45.24
	0.1	**99.55**	**99.54**	**99.56**	**99.48**	**99.39**	**99.24**	97.69	94.89	92.23	89.18	86.02	79.45	68.29
	0.2	**99.38**	**99.60**	**99.64**	**99.54**	**99.43**	**99.23**	98.91	98.32	97.73	97.04	95.66	91.59	82.01
α	**0.5**	**99.34**	**99.24**	**99.41**	**99.55**	**99.44**	**99.21**	98.82	98.44	98.10	97.70	97.15	95.55	91.81
	1	97.72	98.16	98.58	**99.42**	**99.42**	**99.18**	98.66	98.10	97.71	97.41	96.95	95.64	92.30
	1.5	95.22	95.68	96.33	**99.33**	**99.32**	**99.12**	98.73	98.22	97.76	97.45	97.04	95.89	92.61
	2	94.38	94.58	94.92	98.44	**99.20**	**99.13**	98.82	98.35	98.04	97.52	97.13	96.06	93.22
	3	94.10	94.12	94.09	96.84	98.45	98.87	98.79	98.53	98.18	97.83	97.27	96.14	93.94
	5	93.93	94.01	93.96	94.02	96.87	98.13	98.65	98.49	98.32	98.18	97.63	96.77	94.85
	10	93.88	93.87	93.86	93.85	93.84	96.48	97.94	98.36	98.27	98.15	97.93	97.45	95.98

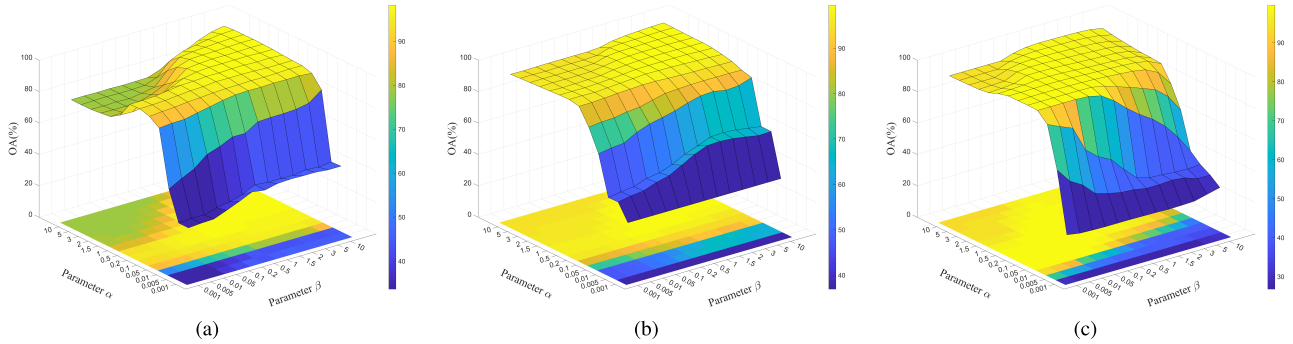
bold underline fonts denote the overall classification higher than 99%.

According to these observations, we could see that both the tensor $l_{2,2,1}$ norm and the position-based Laplacian graph were qualified to handle hyperspectral images. The tensor $l_{2,2,1}$ norm constrained the frontal slice sparsity and ensured the smoothness of local areas. The position-based Laplacian graph was efficient to relieve the impact of the residual noise remaining in the low-rank image by preserving local structures. The $l_{2,2,1}$ norm and the position-based graph cooperated to guarantee the good performance of the proposed LPGTRPCA.

D. Tuning of Regularization Parameters α and β

In this part, we studied the tuning of the regularization parameters α and β. Both of them were set to 0.001, 0.005,

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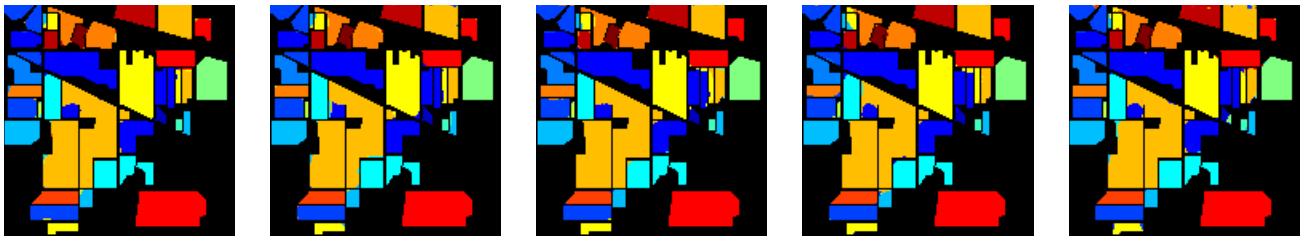
(a) (b) (c)

Fig. 17. OA (%) of LPGTRPCA with different values of regularization parameters α and β on (a) Indian Pines, (b) Pavia University, and (c) Salinas.

TABLE XII
CLASSIFICATION RESULTS OF LPGTRPCA WITH DIFFERENT r'S ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS

Indian Pines						Pavia University						Salinas					
r	1	2	3	4	5	r	1	2	3	4	5	r	1	2	3	4	5
AA(%)	***96.21**	96.10	96.04	95.91	95.71	AA(%)	***98.85**	98.49	98.32	98.38	98.31	AA(%)	99.46	***99.56**	99.53	99.51	99.46
OA(%)	***98.22**	97.99	97.97	97.88	97.55	OA(%)	***99.52**	99.41	99.29	99.31	99.19	OA(%)	99.64	***99.71**	99.69	99.67	99.64
KC(%)	***97.97**	97.71	97.68	97.58	97.21	KC(%)	***99.41**	99.26	99.11	99.14	99.00	KC(%)	99.60	***99.68**	99.65	99.63	99.60

*** and bold fonts denote the best among the group.**



(a) (b) (c) (d) (e)

Fig. 18. Classification maps of LPGTRPCA with different r's on Indian Pines. (a) $r = 1$. (b) $r = 2$. (c) $r = 3$. (d) $r = 4$. (e) $r = 5$.

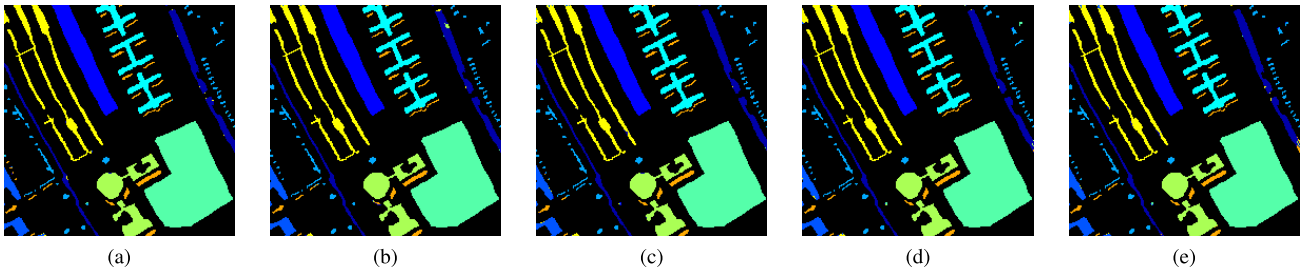
0.01, 0.05, 0.1, 0.2, 0.5, 1, 1.5, 2, 3, 5, and 10. The radius r of the locality preserving graph was fixed at 1. The OA of LPGTRPCA with different parameters α and β is listed in Tables IX–XI and illustrated in Fig. 17. On the Indian Pines datasets, as shown in Table IX, LPGTRPCA got the highest OA (98.22%) with $\alpha = 1$ and $\beta = 3$, and $\alpha = 1.5$ and $\beta = 5$. Values of α and β leading to OA greater than 97.00% were in the intervals [0.2, 10] and [0.2, 10], respectively. When β was fixed, OA first increased and then decreased as α increased. For example, when $\beta = 1$, OA steadily increased from 45.43% to 98.10% as α changing from 0.001 to 1 and decreased from 98.10% to 81.46% as α increasing from 1 to 10. When α was fixed and β changed from 0.001 to 10, OA also first increased and then decreased. On Pavia University, as shown in Table X, LPGTRPCA got the highest OA (99.52%) with $\alpha = 3$ and $\beta = 3$. The $\alpha \in [0.5, 10]$ and the $\beta \in [0.5, 10]$ lead to good OA (higher than 99.00%). On the Salinas dataset, as shown in Table XI, LPGTRPCA obtained the best OA (99.64%) with $\alpha = 0.2$ and $\beta = 0.01$. The values of α and β leading to OA greater than 99.00% were in the intervals [0.05, 0.5] and [0.001, 0.2], respectively. Similar to the case on Indian Pines, with either α or β fixed, OA first increased and then decreased as the other one increased on the Pavia University and Salinas datasets.

Comparing Tables IX–XI and Fig. 17(a)–(c), we have two observations. First, on all datasets, values of α in the interval [0.5, 2] led to a high OA, while a too large or small value led to poor classification results. Second, these three datasets required different values of β for a high OA. Good values for Indian Pines and Pavia University were obviously greater than those for Salinas. This is because Indian Pines and Pavia University have more spare noise than Salinas, so a larger β is needed to enhance the locality-preserving ability of the graph regularization term, to relieve the impact of noise remaining in the low-rank image. Based on these observations, the optimal value of α can be set in the interval [0.5, 2], and the value of β should consider the noise levels of hyperspectral images, i.e., the greater the noise the larger β should be.

E. Influence of Radius r of the Position-Based Graph

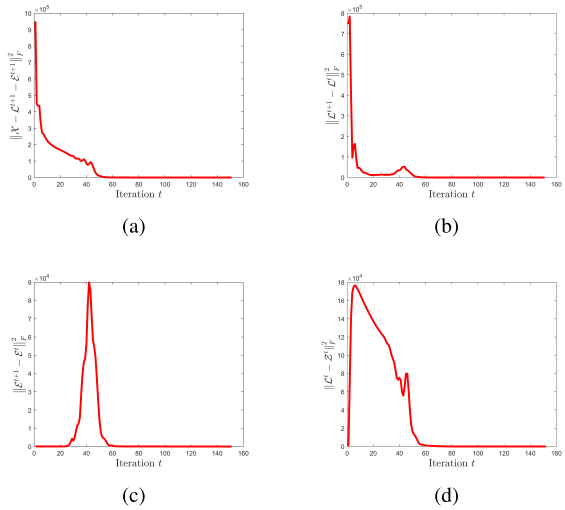
We set the value of r to 1, 2, 3, 4, and 5 to analyze its effect on the position-based Laplacian graph. Table XII listed

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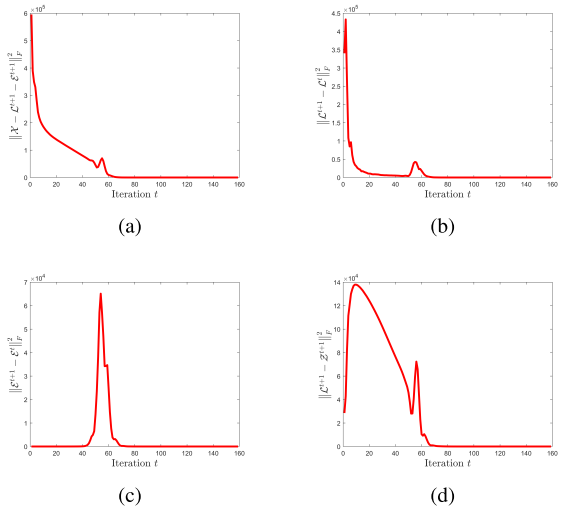
(a) (b) (c) (d) (e)

Fig. 19. Classification maps of LPGTRPCA with different r's on Pavia University. (a) $r = 1$. (b) $r = 2$. (c) $r = 3$. (d) $r = 4$. (e) $r = 5$.



(a) (b) (c) (d)

Fig. 20. Convergence curves of LPGTRPCA on Indian Pines. (a) $\|\mathcal{X}-\mathcal{L}^{t+1}-\mathcal{E}^{t+1}\|_F^2$. (b) $\|\mathcal{L}^{t+1}-\mathcal{L}^t\|_F^2$. (c) $\|\mathcal{E}^{t+1}-\mathcal{E}^t\|_F^2$. (d) $\|\mathcal{L}^t-\mathcal{Z}^t\|_F^2$.



(a) (b) (c) (d)

Fig. 21. Convergence curves of LPGTRPCA on Pavia University. (a) $\|\mathcal{X}-\mathcal{L}^{t+1}-\mathcal{E}^{t+1}\|_F^2$. (b) $\|\mathcal{L}^{t+1}-\mathcal{L}^t\|_F^2$. (c) $\|\mathcal{E}^{t+1}-\mathcal{E}^t\|_F^2$. (d) $\|\mathcal{L}^t-\mathcal{Z}^t\|_F^2$.

the classification results of LPGTRPCA with different values of r on the Indian Pines, Pavia University, and Salinas datasets, and Figs. 18 and 19 showed the classification maps on Indian Pines and Pavia University.

According to Table XII, the performance of LPGTRPCA on Indian Pines and Pavia University datasets deteriorated slightly with the increase in r. For example, on Indian Pines, the OA values obtained by LPGTRPCA were 98.22%, 97.99%, 97.97%, 97.88%, and 97.55%, for $r = 1, 2, \ldots, 5$, respectively. Differently, the performance of LPGTRPCA on Salinas was insensitive to r. For instance, the best OA (99.71%), for $r = 2$, was just 0.07% greater than the lowest OA (99.64%), achieved with $r = 1, 5$. An appropriate r could be chosen according to the properties of a hyperspectral image. Specifically, a small r is suitable for the images with complex local structures, while a larger r should be assigned to the images with simple land cover types.

F. Convergence Study

The convergence of LPGTRPCA was demonstrated in this part. Fig. 20 showed the convergence curves of LPGTRPCA on the Indian Pines dataset, with $\alpha = 1$, $\beta = 3$, and $r = 1$. For each subfigure, the X-axis denotes the number of iterations, and the Y-axis denotes the criterion value. As seen in Fig. 20(a), the value of $\|\mathcal{X} - \mathcal{L}^{t+1} - \mathcal{E}^{t+1}\|_F^2$ decreased sharply with iteration and finally approached zero. This meant the low-rank part $\mathcal{L}$ and the sparse noise part $\mathcal{E}$ obtained by our method could sufficiently recover the original hyperspectral image. In addition, according to Fig. 20(b)–(d), the low-rank part $\mathcal{L}$, the sparse noise part $\mathcal{E}$, and the auxiliary variable $\mathcal{Z}$ all became stable after dozens of iterations. We can see from Fig. 20 that LPGTRPCA converged after about 60 iterations and satisfied the stop criteria after 150 iterations. Similar observations were seen on the Pavia University and Salinas datasets. Fig. 21 showed the convergence curves of LPGTRPCA on the Pavia University, with $\alpha = 3$, $\beta = 3$, and $r = 1$. Fig. 22 showed the convergence curves of LPGTRPCA on the Salinas, with $\alpha = 0.2$, $\beta = 0.01$, and $r = 1$. In conclusion, LPGTRPCA converged on all three datasets.

G. Discussion

In the experiments, LPGTRPCA outperformed various state-of-the-art tensor-based RPCA methods on three real-world hyperspectral images in terms of AA, OA, and KC. This proved the superiority of the proposed method. The ablation study showed that the new defined $l_{2,2,1}$ norm could not only extract sparse noise from hyperspectral images as efficiently as $l_{2,1,2}$ norm but also lead to the smoother locality, and

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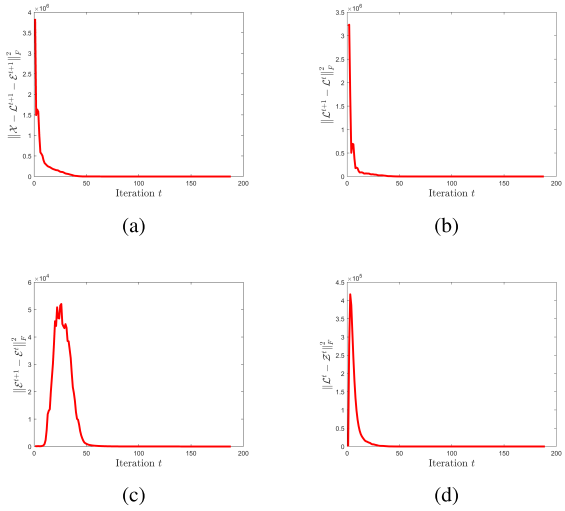


Fig. 22. Convergence curves of LPGTRPCA on Salinas. (a) $\|\mathcal{X} - \mathcal{L}^{t+1} - \mathcal{E}^{t+1}\|_F^2$. (b) $\|\mathcal{L}^{t+1} - \mathcal{L}^t\|_F^2$. (c) $\|\mathcal{E}^{t+1} - \mathcal{E}^t\|_F^2$. (d) $\|\mathcal{L}^t - \mathcal{Z}^t\|_F^2$.

the position-based Laplacian graph significantly relieved the impact of the residual noise. This demonstrated the innovation of the research. Besides, the analysis of the effect of tradeoff parameters α and β and radius r of position-based graph reflected that the suitable values of these parameters could be easily set. Finally, the convergence of LPGTRPCA was illustrated to show its stability and reliability.

V. Conclusion

In this article, we proposed LPGTRPCA to achieve robust classification results for hyperspectral images. LPGTRPCA decomposed a raw hyperspectral image into a noise part with frontal slice sparsity constrained by the tensor $l_{2,2,1}$ norm and a low-rank part with locality preserved by the position-based Laplacian graph. This method was solved by ADMM, and explicit solutions were provided. The experiments conducted on three real-world hyperspectral images demonstrated the reliability and innovation of LPGTRPCA.

It is noteworthy that there is room to improve the LPGTRPCA method. For example, to preserve local structures of the low-rank image more precisely, the shape of the window that decides the neighbor set of each pixel could be automatically set according to local properties, instead of just setting it to square. In addition, if the tradeoff parameters α and β could be adaptive-adjusted, LPGTRPCA would get more stable results. What is more, we will consider combining the proposed method with deep learning methods [56]–[58] to further improve its performance. These problems will be figured out in our future work.

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Yingxu Wang received the B.S. degree in electronics and information engineering from Beihang University, Beijing, China, in 2015, and the M.S. degree in computer science and technology from the University of Jinan, Jinan, China, in 2019. He is pursuing the Ph.D. degree with the Department of Computer and Information Science, University of Macau, Macau, China.

His research interests include machine learning, image segmentation, and their applications.



Tianjun Li received the B.S. degree in computer and information science from Zhengzhou University, Zhengzhou, China, in 2013, and the M.S. and Ph.D. degrees in computer and information science from the University of Macau, Macau, China, in 2016 and 2020, respectively.

He is a Post-Doctoral Fellow with the School of Computer Science and Engineering, South China University of Technology, Guangzhou, China. His research interests include bioinformatics, data mining, machine learning, and optimization methods.

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Long Chen (Member, IEEE) received the B.S. degree in information sciences from Peking University, Beijing, China, in 2000, the M.S.E. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, in 2003, the M.S. degree in computer engineering from the University of Alberta, Edmonton, AB, Canada, in 2005, and the Ph.D. degree in electrical engineering from The University of Texas at San Antonio, San Antonio, TX, USA, in 2010.

From 2010 to 2011, he was a Post-Doctoral Fellow with The University of Texas at San Antonio. He is an Associate Professor with the Department of Computer and Information Science, University of Macau, Macau, China. His research interests include computational intelligence, Bayesian methods, and other machine learning techniques and their applications.



Yufeng Yu received the Ph.D. degree in statistics from Sun Yat-sen University, Guangzhou, China, in 2017.

From 2016 to 2017, he was a Visiting Scholar with the Lane Department of Computer Science and Electrical Engineering, West Virginia University, Morgantown, WV, USA. From 2017 to 2018, he was a Senior Research Associate with the Department of Electronic Engineering, City University of Hong Kong, Hong Kong, China. He is a Post-Doctoral Fellow with the Department of Computer and Information Science, University of Macau, Macau, China. His research interests include image processing, statistical optimization, pattern recognition, and machine learning.



Yinping Zhao received the B.S. and M.S. degrees from the School of Electrical Engineering and Automation, Hefei University of Technology, Hefei, China, in 2012 and 2015, respectively, and the Ph.D. degree in computer and information science from the University of Macau, Macau, China, in 2021.

She is an Associate Professor with the School of Software, Northwestern Polytechnical University, Xi'an, China. Her research interests include artificial intelligence, machine learning, pattern recognition, and image processing.



Jin Zhou (Member, IEEE) received the B.S. and M.S. degree in software engineering from Shandong University, Jinan, China, in 1998 and 2001, respectively, and the Ph.D. degree in software engineering from the University of Macau, Macau, China, in 2014.

He is a Professor with the Shandong Provincial Key Laboratory of Network Based Intelligent Computing, University of Jinan, Jinan, China. His research interests include intelligent transportation systems, machine learning techniques, and their applications.

Dr. Zhou has served in many international conferences as the workshop/special session chair.

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